

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 1.1**

RELATED SUPPORT

- **TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1**

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Key Project Information

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Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

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KEY PROJECT INFORMATION

GS ID of Project	VPA GS ID: 11259 PoA GS ID: 11189
Title of Project	VPA Title: Improved Cookstove and Safe Water Programme – Nigeria – VPA 01 PoA Title: Improved Cookstove and Safe Water Programme
Time of First Submission Date	30/07/2021
Date of Design Certification	02/02/2022
Version number of the VPA-DD	5.0
Completion date of version	18/05/2022
Coordinating/managing entity	Impact Carbon LLC
VPA Implementer (s)	Impact Water LLC
Project Participants and any communities involved	Impact Carbon LLC Impact Water LLC
Host Country (ies)	Nigeria
GS ID and Title of applicable Design Certified VPA	--
GS ID and Title of applicable Performance Certified VPA	--
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale
Other Requirements applied	GS4GG Principles & Requirements Version 1.2 GHG Emissions Reduction & Sequestration Product Requirements Version 2.0 Programme of Activity Requirements Version 1.2
Methodology (ies) applied and version number	GS Approved Methodology "Emission reductions from Safe Drinking Water Supply" Version 1.0 – 03/5/2021
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Amount of CO ₂ e reduced by the project	59,312	tCO ₂ e (VER)
SDG: 1 No Poverty 1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance	1.4.1 Proportion of population living in households with access to basic services Indicator: Total number of premises (Schools / institutions) with at least one WPS distributed / installed under the project	650	Number
SDG:3 Good Health and Well Being 3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	3.9.2 Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and Hygiene for All (WASH) services) Indicator: % sampled Schools / institutions reporting reduction in incidence of diarrhoea and water borne diseases etc. after shifting to the project WPS	90%	Percentage
6 Clean Water and sanitation 6.1 By 2030, achieve universal and equitable access to safe and affordable drinking water for all	6.1.1 Proportion of population using safely managed drinking water services Indicator: % of WPS distributed/installed providing safe drinking water quality	90%	Percentage
7 Affordable and Clean Energy 7.1 By 2030, ensure universal access to affordable, reliable and modern energy services	7.1.2 Proportion of population with primary reliance on clean fuels and technology Indicator: % Schools / institutions reporting an operational WPS in project	90%	Percentage
8 Decent Work and Economic Growth 8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities Indicator: Number of male / females employment created by project	19	Number

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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The stated goal of proposed VPA is widespread dissemination of low GHG water purification technologies (WPS) to schools and other institutions in Nigeria. The technologies implemented under the VPA are further described in section A.3 below. The VPA will use carbon finance to support local partners engaged in operation, sales and distribution and maintenance of various WPS technologies.

The VPA will reduce GHG emissions by reducing or replacing the use of non-renewable biomass / fossil fuel for boiling water to purify it for drinking purposes.

In Nigeria, only 11.7% of the population have access to piped water, with only 3.5% population having access to piped water within their homes¹. In Nigeria, 80.6% of population use solid fuel for cooking purposes. In rural, 83.0% population use firewood while 3.0% use charcoal and in urban, 42.1% of population use firewood and 11.8% of population use charcoal for cooking². 90 million Nigerians, cook/boil water on traditional stoves / "three-stone fire" cooking technology³. The baseline study conducted to assess the baseline scenario in schools and institutions in Nigeria is stated in section B.4 below.

Prevailing Policies:

President of Nigeria launched the National Action Plan (NAP), a 13-year strategy for the Revitalization of Nigeria's Water Supply, Sanitation, and Hygiene (WASH) Sector⁴. The goal of the Revitalization Strategy is to ensure that all Nigerians have access to sustainable and safely managed WASH services by 2030, in compliance with the Sustainable Development Goals (SDG) for Water (Goal 6.1) and Sanitation (Goal 6.2).

The project WPS will result in elimination/reduction of non-renewable biomass / fossil fuels usage for water boiling. This will result in significant improvement in indoor air pollution associated with use of solid biomass / fossil fuel based inefficient cookstoves used for water boiling which has a direct correlation with respiratory illness and mortality rates, especially among school children in the project beneficiaries.

In this VPA, the VPA implementer is Impact Water LLC, and the CME is Impact Carbon LLC. The VPA Implementer shall operate the VPA as per the CME management system and provide the CME with information required to include the project activity under the PoA, perform monitoring and verification of the activity, and renew the crediting period, besides other responsibilities. Beneficiaries receiving the project technologies under the VPA will agree to the terms of the PoA and will cede all rights to any VERs resulting from the VPA to CME (Impact Carbon) or CPA Implementer (Impact Water), as applicable.

¹ Table 2.1.1, Nigeria Demographic and Health Survey 2018, Report, published in October 2019

² Table CH.12, Nigeria Multiple Indicator Cluster Survey 2016-17, published in February 2018,

³ Page 88, The Truth About Cooking Landscape Analysis, Global Alliance for Clean Cookstoves (GACC), October 2016,

⁴ <https://www.wateraid.org/ng/sites/g/files/jkxoof381/files/nigerias-national-action-plan-for-the-revitalization-of-the-wash-sector.pdf>

Target Group

The VPA's target groups are schools and institutions, within Nigeria that purify water, or would have purified water (suppressed demand), through boiling using non-renewable biomass / fossil fuel.

Distribution Method

The VPA Implementer will either distribute technologies directly to the target groups (i.e., schools or institutions) in Nigeria or work with local distributors to distribute the WPS technologies to the target group.

Technology

The technology distributed / sold under the VPA are low-GHG water purification technologies (refer section A.3 below for details).

Record Keeping System

Upon installation/distribution of the project WPS units, and associated accessories as applicable, following data would be collected for the project unit⁵:

- a. Address and contact details (name and phone number if available) of the beneficiary
- b. Date of distribution / installation
- c. WPS Technology type (model, make etc.)
- d. Unique ID number / TAG number / Batch number, as applicable.

The record keeping database will also be used to record the results of all monitoring activity, with all data collected to be kept for at least two years after the end of crediting period or the last issuance of VERs for the project activity, whichever is later. The data will be stored electronically in the database.

The VPA follows the GS Methodology "Emission reductions from Safe Drinking Water Supply" Version 1.0 – 03/5/2021.

Contribution to Sustainable Development

The VPA contributes to sustainable development by increasing accessibility to low GHG safe drinking water technologies among the target groups within the VPA boundary. In addition to increasing accessibility, the activity implemented under the VPA will reduce the use of fossil fuels and non-renewable biomass thereby decreasing GHG emissions within the VPA boundary and reducing fuel expenditures. It will also provide health benefits, such as the reduction of waterborne disease by increasing access to safe drinking water and improved indoor air quality by reducing IAP emissions. Refer table 1 above for anticipated SD benefits from the project.

A.1.1. Eligibility of the project under approved PoA

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⁵ It may not be possible to collect beneficiary details for all WPS distributed under a VPA. However, the beneficiary details should be collected for as many beneficiaries as commensurate with representative sampling, i.e. it should not be less than 10 times the survey and field test sample size.

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/ Supporting evidence
1.	Technology	For WPS Project technology performance level (IWT): It shall be demonstrated based on report of laboratory testing or official notification that the project technology or equipment achieves either - Compliance with the national standard or guideline for household drinking water treatment technology, or - The performance target classification 3-star or 2-star level, meaning "Comprehensive Protection," as per the WHO International Scheme to Evaluate Household Water Treatment Technologies (World Health Organization, 2011)	The WPS technologies under the VPA are in compliance with the national standards. Supporting Evidence: Technical specifications / official notification from approved agencies for Chlorination (UltraTAB / UltraFLO).
2.	Location	Each VPA will be located within the physical/geographical boundary of the PoA	Refer section A.2 (location of project) of VPA-DD for location of the VPA. The WPS Distribution / Installation Database lists all systems as installed within Nigeria.
3.	Additionality	Each VPA will satisfy the criteria for demonstrating additionality through one of the following options: Option 1: As per Activity Requirement: As per GS4GG Community services activity requirements, Version 1.2, Para 4.1.9, Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of design certification: (a) Positive list (Annex B of this document) (b) Projects located in LDC, SIDS, LLDC (c) Microscale projects Option 2: CDM Tool Para 12 and 13 of Tool 19 (version 9.0); or	The project is additional as per CSA Annex B. Refer section B.5 below for details.

		• Para 10/Figure 1 of Tool 21 (Version 13.1); or • Para 11 of Tool 21 (Version 13.1)	
4.	De-Bundling	As per GS4GG Programme of activities requirements section 10.1.1, de-bundling provisions do not apply to Voluntary PoAs.	Not required
5.	Double Counting	Each VPA will utilize identifiers for every appliance under the PoA to show that the appliance belongs to that specific PoA. The unique identifier will be designating each appliance as part of the PoA, and CME master distribution/installation records will ensure each sale is credited under only a single VPA.	CME maintains a VPA distribution database listing all distributed WPS included in the VPA. Each WPS unit bears a unique product ID to avoid double counting. Supporting Evidence: WPS Distribution / Installation Database
6.	History	VPAs are neither registered as project activities with other offset Schemes, included in other registered PoAs, nor the project activities that have been deregistered, unless transitioning to GS from other standard.	The systems included in the VPA have claimed credits under CDM PoA 9948 previously (till 31 Dec 2020). The systems included in the VPA start their crediting from 01 Jan 2021 onwards which is exclusive of any credits claimed previously by these systems in CDM PoA 9948. Supporting Document: CME declaration
7.	Start Date	Each VPA will prove that the start date of the VPA is on or after the start date of the PoA, or state that the VPA is claiming credits retroactively. As the project involves distribution / installation of WPS (distributed technology) the start date is the date of implementation of the first unit under the project.	The start date of the VPA is the 12/01/2018.. Supporting Evidence: First WPS sales receipt / installation log / delivery note etc.
8.	Crediting Period	Each VPA will have a renewable crediting period	The VPA has a renewable crediting period. Refer section C (duration and crediting period) of VPA-DD for crediting period
9.	Public Funding	Each VPA will confirm that it does not results in diversion of Official Development Assistance (ODA). Declaration by CME that the VPA does not involve any funding that results in diversion of ODA.	Refer section A.5 (funding source of project) below. Supporting Documents: ODA Declaration by CME /VPA Implementer.
10.	CME Approval	Each VPA will prove it has received the approval of the CME of the PoA.	The VPA has been approved by CME for inclusion in the PoA. Supporting Document: CME approval letter

11. Methodology	VPA will apply the GS methodology: "Emission Reductions from safe drinking water supply" Version 1.0 and adhere to all applicability conditions and other requirements of the methodology	Refer section B.1 (reference of approved methodologies) below where the same has been confirmed.
12. Target Group	The target group of the PoA, and each included VPA, are schools and/or institutions:	Refer Distribution database confirming the target group as schools and other institutions. Supporting Document: Distribution / Installation database
13. Sampling	VPA will adhere to the sampling requirements stipulated in "Standard for Sampling and surveys for CDM project activities and programmes of activities" version 09.0.	Refer section B.7.2 (sampling plan) of VPA-DD below.
14. Stakeholder Consultation and Environmental Analysis	VPA will conduct a Local Stakeholder Consultation / SFR, as applicable and adhere to the Environmental Impact Analysis requirements of the host country	VPA Local stakeholder consultation/SFR has been conducted. according to Nigeria Environmental Impact Assessment (EIA) Decree 1992, the proposed project falls outside the project category for which environmental impact assessment is mandatory hence EIA is not deemed required for WPS technologies in host country Nigeria Supporting Document: Consultation Report
15. VER Ownership	Each VPA will assure ownership of the VERs is secured by the CME	Refer purchase orders / sales receipts which transfers the rights of VERs from beneficiary to CME or VPA Implementer. Also, this has been communicated to stakeholders during SFR.
16. Meth threshold	Each WPS VPA will ensure that it will meet the small-scale thresholds and remain within those thresholds throughout the crediting period	The VPA is limited to small-scale threshold (60k VERs per annum). Supporting Document: ER spreadsheet
17. SDG outcome assessment	The monitoring plan for SDG shall include: 1. reduction in smoke, PM, soot emissions after shifting to the project ICS 2. reduction in incidence of diarrhoea and water borne diseases etc. after shifting to the project WPS 3. Number of ICS/WPS distributed and operating. 4. Total Number of WPS distributed/installed under the project and % of WPS	SDG monitoring plan includes all details. Refer Section B.7.1 of the VPA-DD

	distributed/installed provide safe drinking water quality	
	5. Number of male/female persons hired.	
	Fuel savings reported by users in the PoA after shifting to ICS	
18. Safeguarding Principles	The CME shall conduct the Safeguarding Principles Assessment as per the Safeguarding Principles & Requirements at the VPA/CPA equivalent level.	Refer Section E below for the SPA at the VPA level.
19. Retroactive VPAs	Retroactive VPAs that are submitted to GS /SustainCert at a date later than one year from the VPA start date shall not be eligible for Gold Standard Certification.	The VPA being retroactive has already concluded GS / SustainCert preliminary review, successfully, on 30/11/2021. Thus, the VPA is deemed eligible for Gold Standard Certification.
20. CER Labelling	Projects in other standards seeking labelling of CERs under GS4GG shall demonstrate compliance with section 2.0, Annex B of GHG Emissions Reduction and Sequestration Product Requirements	Not Applicable, CER labelling is not envisaged for this VPA.
21. Conditions to be met in multi-country PoAs	Not applicable. Although the PoA is multi-country, a VPA shall remain limited to a singular country in the PoA and shall not include more than one country in the VPA boundary.	Although the PoA is multi-country, project boundary of the VPA-DD is Nigeria. Refer section A.2 of the VPA-DD.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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CME (Impact Carbon LLC) or VPA Implementer (Impact Water LLC) have full and uncontested legal ownership of GS VERs (SDG 13) that are generated from use of WPS products distributed under the VPA (VERs rights). The project beneficiaries agree to transfer the ownership of GS VERs to CME (Impact Carbon LLC) or VPA Implementer (Impact Water LLC) at the time of purchase / distribution of product.

A.2. Location of project

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Host Country: Nigeria

Region/State/Province etc.: All regions of Nigeria

City/Town/Community etc.: All cities and towns in Nigeria

PoA Title: Improved Cookstove and Safe Water Programme

VPA Title: Improved Cookstove and Safe Water Programme – Nigeria – VPA 01
PoA GS ID: 11189
VPA GS ID: 11259

Physical/Geographical location:

Nigeria lies between 4° 16' and 13° 53' north latitude and between 2°40' and 14° 41' east longitude and has a land area of 924,000 sq. km, one of the largest in Africa.

The map of Nigeria, for reference purpose, is as follows:



A.3. Technologies and/or measures

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Low GHG Water Purification Technologies

The project intends to make WPS products available to the schools and institutions in Nigeria. The project proponent will continually assess technology options and partnerships, with the goal of providing the high performance, affordable and appropriate technologies to beneficiaries in Nigeria.

The technologies/measures employed under the VPA are low GHG water purification technology involving point-of-use (POU) water treatment systems for schools and institutional applications. These will replace the current practice of boiling water with fossil fuel and/or non-renewable biomass in project schools which is the baseline scenario.

Several type of water purification technologies are distributed under the VPA, that achieves compliance with the national standard or guideline for drinking water treatment technology. The example water purification systems and expected specifications, to be distributed initially through the project include the following:

WPS Models	Technology Type	Lifetime ⁶	Fixed or Portable	Removal of E. coli	Minimum Watt / Voltage
UltraFLO	Chemical	Expiry: 5 year Capacity: 340,000 ltrs	Fixed	99 (2-log)	Not applicable
UltraTAB	Chemical	Expiry: 5 years Capacity: Big pack: 48,000 ltrs Small pack: 10,000 ltrs	Portable	99 (2-log)	Not applicable

All technologies included in the VPA are suitable within the context of local water consumption and cooking practices.

Along with the WPS that are currently being implemented, other low carbon water purification system may also be included in the VPA later during the crediting period provided they meet the requirements of applied methodology and GS4GG requirements.

A.4. Scale of the project

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The project is a small-scale project.

A.5. Funding sources of project

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The project activity does not receive any public funding from Annex I countries and diversion of official development assistance (ODA) is not involved in the project activity.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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GS Standard Approved Methodology:

The first crediting period will follow GS approved methodology i.e., "Emission reductions from Safe Drinking Water Supply" Version 1.0 – 03/5/2021 for baseline and monitoring requirements. The applied methodology can be access at the following link:

<https://www.goldstandard.org/project-developers/standard-documents>

Methodological Tools:

CDM Tool 30: Calculation of the fraction of non-renewable biomass

⁶ Estimated lifetime from manufacturer specification sheet. This expiry is the valid lifetime of the chlorine tablet / cartridge within which it is deemed to be effective. The system lifetime is deemed greater than 20 years as other than the tablet, cartridge it consists of tanks and piped connections (for UltraFLO) which are highly durable and long lasting and can be easily replaced.

Guidelines:

N/A

Standardized baseline:

Not applicable

Additionality Tool:

Annex B of CSA requirements

B.2. Applicability of methodology (ies)

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The project applied methodology "Emission reductions from Safe Drinking Water Supply Version 1.0 – 03/5/2021". The following conditions apply to VPAs under this methodology:

The following conditions apply to VPAs under this methodology:

Applied Methodology Applicability Criteria	Justification
This methodology is applicable to project activities that introduce a new, or rehabilitate an existing, zero-emission ⁷ or low-emission technology to supply safe drinking water.	The VPA involves distribution of Low GHG Emission Water Purification Systems ("WPS") in schools and Institutional premises ⁸ that would have boiled water in the baseline.
Technologies include household water treatment technologies (HWT), Institutional water treatment technologies (IWT), Community level water treatment technologies (CWT) and community water supply technologies (CWS).	The Project technology includes Institutional water treatment technologies (IWT).
Under this Methodology, a project's objectives are to reduce or avoid greenhouse gas emissions from boiling unsafe drinking water in the baseline, and to supply drinking water that is safe for consumption when it enters the project households or institutional premises. When the drinking water is treated in the household or institution (HWT or IWT), then the water supplied from the treatment technology should be safe. When the water is supplied or	The VPA involves distribution/installation of low greenhouse gas, Institutional water treatment technologies (IWT). These are point of use systems, located in the school/institution premises. These systems avoid GHG emissions from boiling unsafe drinking water in the baseline. Ex-post monitoring shall be conducted on sampling basis to demonstrate that the project systems deliver safe water.

⁷ Zero emission technology refers here to emissions generated by technologies once installed within the targeted premises and operational – it does not refer to life cycle emissions such as upstream emissions associated with the production or delivery of the technology.

⁸ e.g., residential institutional, industrial, or commercial facilities.

retrieved from a CWT or CWS directly to the premises of the household or institution, then the water entering the end-user premises should be safe.	Emission reductions shall be accounted accordingly.
The methodology is applicable under the following conditions:	Please refer below for justifications:
a. Eligible household water treatment technologies (HWT), institutional water treatment technologies (IWT), and community level water treatment technologies (CWT) include bleach/chlorine, water filter (ceramic, sand, composite, membrane, etc.), UV disinfection, etc.	Examples of WPS technologies to be included in the VPA, but not limited to, are as follows: Chemical disinfection (chlorine) – UltraTAB/UltraFLO
b. Eligible community water supply technologies (CWS) include new installation of new borehole hand-pumps, borehole hand-pumps rehabilitation, solar powered drinking water pumps, etc. Water pumps powered by fossil-fuel engines are not eligible, with the exception of backup fossil-fuel engines that are used for no more than 10% of operating hours (parameter SWDS 33).	Not applicable, as VPA involves distribution/installation of low greenhouse gas point of use Institutional water treatment technologies (IWT).
c. All projects involving CWT and CWS technologies must also include ongoing maintenance and repair of the project technology.	Not applicable, no CWT or CWS technologies are involved.
d. Where the project involves the rehabilitation of an existing technology, the project developer shall provide evidence that the existing technology is non-operational and that there is no planned maintenance or repair for at least 3 months after the date it became non-operational.	Not applicable, rehabilitation of existing technology is not envisaged in the VPA.
e. This methodology allows for project activities to include safe water treatment and/or supply technologies implemented for end-users in households, and/or commercial premises such as shops or institutional premises including half or full day/boarding schools, prisons, army camps & refugee camps.	The VPA involves distribution/installation of low GHG, Institutional water treatment technologies (IWT) in schools and other Institutions in Nigeria.

f. In cases where the safe water is retrieved at the CWT or CWS location, the water in its improved form shall be available within a distance of 1 km or less from the end-users, as demonstrated by satellite imaging or GPS coordinates⁹ of each CWT or CWS location. Alternatively, as a proxy, a total collection time of 30 minutes or less for a round trip, including queuing, using the travel modes of walking or pedaling may be demonstrated.	Not applicable, no CWT or CWS technologies are involved.
g. Project technology performance level (HWT and IWT): It shall be demonstrated based on report of laboratory testing¹⁰ or official notification¹¹ that the project technology or equipment achieves either (i) the performance target classification 3-star or 2-star level, meaning "Comprehensive Protection," as per the WHO International Scheme to Evaluate Household Water Treatment Technologies¹² (World Health Organization, 2011) or (ii) compliance with the national standard or guideline¹³ for household drinking water treatment technology; if no national guideline or standard is available, then the project technology shall comply with the	The project technologies meet compliance with the national standard/guidelines. Please refer the confirmation letter from local agency.

⁹ Acceptable formats for GPS coordinates include DMS (degrees, minutes and seconds), DMM (degrees and decimal minutes), and DD (decimal degrees).

¹⁰ The testing should be undertaken under conditions that are representative of the operation conditions of the project site(s) including feedwater.

¹¹ For example, notifications from the national authority on health.

¹² International Scheme to Evaluate Household Water Treatment Technologies, WHO Refer to the list of products tested by WHO https://www.who.int/water_sanitation_health/waterquality/household/hwts-products-evaluated/en/

¹³ The national standard or guideline shall be based on laboratory efficacy testing that, at a minimum, includes quantitative microbial measures of pre- and post-treatment challenge waters that are representative of potential drinking water sources, and that includes measured reductions based on at least one pathogen class (bacteria, viruses, protozoa). "Challenge water" is synonymous with "test water". This is the experimental water that has been spiked with microbes (a "microbial challenge") in order to demonstrate the potential for the technology to reduce microbes.

WHO International Scheme requirements as per (i).	
h. Project technology performance level (CWT and CWS): For each individual CWT or CWS, it shall be demonstrated at the start of each crediting period with water quality testing reports that the water directly supplied by the project water technology/source achieves both: i. microbial quality in line with either (i) national standards or guidelines for microbial quality of drinking water, or in the absence of such requirements, (ii) the guideline values for verification of microbial quality from the Guidelines for drinking-water quality (Table 7.10, WHO, 2017) ¹⁴ ; and ii. compliance with (i) national standards or guidelines on priority chemical contamination and physical and aesthetic aspects, or in the absence of such requirements, (ii) international standards or guidelines on priority chemical contamination ¹⁵ and physical and aesthetic aspects. (parameter SWDS 3).	Not applicable, no CWT or CWS technologies are involved.
i. The project must conduct annual water hygiene education campaigns for the end-users.	The project will conduct annual water hygiene education campaigns for the end-users. This has been developed as a monitoring parameter. Refer monitoring plan section below.
Project shall document the national, regional and local regulatory framework for provision of safe drinking water in the project boundary (parameter SDWS 4). The project shall not undermine or conflict with any national, sub-national and local regulations or guidance for safe drinking water supply, operation,	For national, regional and local regulatory framework for provision of safe drinking water in the project boundary refer section A.1 of the VPA-DD. The VPA does not undermine or conflict with any national, sub-national and local regulations or guidance for safe drinking

¹⁴ World Health Organization. (2017). Guidelines for Drinking-water Quality: fourth edition incorporating the first addendum. Geneva: World Health Organization.

¹⁵ At the global level, the priority chemical contaminants are arsenic and fluoride. In absence of relevant national standards, compliance with the WHO guideline values (maximum 10 µg/L and 1.5 mg/L, respectively) shall be demonstrated.

and maintenance, including any tariff requirements.	water supply, operation, and maintenance, including any tariff requirements.
If the expected technical life of project technology (parameter SDWS 7) is shorter than the crediting period, describe measures to ensure that end users are provided replacement systems of comparable quality at the end of the expected technical life (for example, replace with comparable or better technology, retrofit with performance guarantee, etc.). This applies both for new technology and rehabilitated.	The technical lifetime of the systems is not limited. Instead, the system lifetime is based on the lifetime of the cartridge / Tab Pack as applicable. Depending on the type of system installed/distributed, the lifetime is deemed renewed automatically, every time the system is reinforced with a new cartridge / TAB pack Thus, the systems are not constrained with respect to lifetime. The supplies (quantity and date) made are being monitored as part of proposed monitoring plan for this purpose.
All CWT and CWS projects must include ongoing maintenance and repair of the project technology. The PDD must describe the maintenance and repair plan, including the system for logging/documenting of technology operation and maintenance events including periods of downtime ¹³ . The log of operation and maintenance shall be required during the monitoring period to demonstrate project technology operation”.	Not applicable, the VPA does not involve CWT/CWS systems.

B.3. Project boundary

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Project Boundary

The requirements apply to boundary of VPAs under this methodology:

- a) The project boundary is the physical, geographical sites of the project technologies and potentially of the baseline and project fuel collection and production (e.g., charcoal), as well as solid waste and effluents disposal or treatment facilities associated with the fuel processing.

The project boundary for the VPA is comprised of the schools and institutions where the project units are physically located, and the baseline fuel collection and production points, where wood or charcoal would have been produced or collected, in areas surrounding the project schools and institutions.

- b) The region(s) or town(s) within a single country define the target area. The target area provides an outer limit to the project boundary in which the project has a target population.

The target area is Nigeria.

- c) In cases where woody biomass (including charcoal) is the baseline fuel or where the project activity introduces the use of a new biomass feedstock into the project

situation, the fuel production and collection area is the area within which this woody or new biomass is produced, collected and supplied.

The woody biomass fuel is expected to be sourced from within the project boundary or target area, given biomass-based cooking is prevalent in Nigeria as explained in section A.1 above. The project does not involve introduction of new biomass feedstock.

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Emissions from combustion, in baseline	CO ₂	Yes	Important source of emissions
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Important source of emissions
Project scenario	Emissions from combustion, production, and transport in project	CO ₂	No	VPA replaces the use of nonrenewable biomass/fossil fuel, neither does it requires electricity consumption, hence no project emissions are envisaged.
		CH ₄	No	
		N ₂ O	No	

All water purification systems disseminated under the VPA are located within the VPA boundary shown in section A.2. The VPA boundary is within the host country(ies) covered by the PoA, i.e., the boundary of the PoA.

B.4. Establishment and description of baseline scenario

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President of Nigeria launched the National Action Plan (NAP), a 13-year strategy for the Revitalization of Nigeria's Water Supply, Sanitation, and Hygiene (WASH) Sector¹⁶. The goal of the Revitalization Strategy is to ensure that all Nigerians have access to sustainable and safely managed WASH services by 2030, in compliance with the Sustainable Development Goals (SDG) for Water (Goal 6.1) and Sanitation (Goal 6.2).

National Water Resources Bill- 2020, enacted by National assembly of Federal Republic of Nigeria, establishes regulatory framework for the water resource sector in Nigeria and provide framework for the equitable and sustainable development, management, use and conservation of Nigeria's surface and ground water resources. The objective of the bill is to also ensure the citizen's right of access to safe water and basic sanitation.

The establishment of Nigerian Standard for Drinking Water Quality (NSDQW) ensures the safety of the drinking water supplies and the protection of public health.

¹⁶ <https://www.wateraid.org/ng/sites/g/files/jkxoof381/files/nigerias-national-action-plan-for-the-revitalization-of-the-wash-sector.pdf>

Even though there are policy and framework in place in Nigeria related to safe water drinking still in Nigeria, only 11.7% of the population have access to piped water, with only 3.5% population having access to piped water within their homes¹⁷. In Nigeria, 80.6% of population use solid fuel for cooking purposes. In rural, 83.0% population use firewood while 3.0% use charcoal and in urban, 42.1% of population use firewood and 11.8% of population use charcoal for cooking¹⁸. 90 million Nigerians, cook/boil water on traditional stoves / “three-stone fire” cooking technology¹⁹. The baseline study conducted to assess the baseline scenario in schools and institutions also substantiate that access to safe drinking water for the population is still a challenge in Nigeria.

As per section 3.4.1 of the applied GS methodology, the general baseline scenario is that users would have boiled drinking water in the absence of the project activity.

As per section 3.4.3 of the applied methodology, for the case of schools/institutions currently drinking unsafe water because e.g. energy poverty barriers result in less than the minimum required amount of safe drinking water, the principles of suppressed demand are applied and the baseline is set as a proxy technology (water boiling of an adequate quantity of drinking water) based on the standard of living achieved by peers (adequate supply of safe drinking water).

As per section 3.4.2 of applied methodology, the suppressed demand baseline may be applied for institutional end-users, except where the institution is connected to a public distribution network (PDN) that supplies safe drinking water – unless justified that supplied water quality doesn’t meet safe water definition.

Therefore, Institutional end users, connected to PDN, who are consuming unsafe water due to resource constraint are also included in the baseline scenario.

As per section 3.5.1 of the applied GS methodology, the applicable baseline scenario for fuel, technology and end user group will be determined as per guidelines on baseline scenario selection and justification provided in Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC, Version 3.1).

As per applied methodology (Version 1.0) section 4.2.3, the baseline survey should be following these guidelines for minimum sample size:

Group size <300: 30 or population size, whichever is smaller
 Group size 300 to 1,000: 10% of group size
 Group size > 1,000: 100

As per official website of federal ministry of education, Nigeria, there are over 125,000 schools in Nigeria. Therefore, the sample size to considered as per methodology is 100. The CME conducted baseline survey targeting 200 schools spread across Nigeria (spread across 16 counties) out of which surveys were completed for 170 schools successfully.

¹⁷ Table 2.1.1, Nigeria Demographic and Health Survey 2018, Report, published in October 2019

¹⁸ Table CH.12, Nigeria Multiple Indicator Cluster Survey 2016-17, published in February 2018,

¹⁹ Page 88, The Truth About Cooking Landscape Analysis, Global Alliance for Clean Cookstoves (GACC), October 2016,

The section below presents the outcome of the survey. The sources of water in these 170 schools were verified as follows:

Source of drinking water	# of Schools/Institutions	% of Schools/Institutions
Well/borehole	160	94.12%
Surface Water	4	2.35%
piped government	4	2.35%
piped private	2	1.18%

As per Annex 2 of the applied GS methodology, piped water and water from boreholes or protected wells belong to improved sources of drinking water. For the remaining, the water source has been considered as unimproved.

Source of drinking water	# of Schools/Institutions	% of Schools/Institutions
Improved sources	166	97.65%
Unimproved Sources	4	2.35%

Pre-project practices of boiling water, or drinking unsafe water (suppressed demand):

The baseline survey questionnaire, confirmed from the sample of the practice of treating water (either via boiling or other technique) or drinking unsafe water. Based on the same the following was deduced from survey data collected.

Baseline Treatment Method	
Type of treatment method	# of Schools/Institutions
Receiving safe water without treatment	2
Treating/would have treated, water without boiling	6
Treating/would have treated, water via boiling	162
C_b	4.71%

Therefore, C_b (Proportion of project end-users who in the baseline were already using safe water, either from an improved water source, or from a water treatment method other than boiling) is =4.71%.

Baseline Fuels:

The types of baseline fuels reported by sampled school included woodfuel, charcoal and kerosene. The distribution of different fuel types being consumed is as follows:

Baseline Fuel Usage	
Type of fuel consumed	% Schools/Institutions
Woodfuel	58.02%
Charcoal	40.74%
Kerosene	1.23%
Total	100.00%

So solid biomass fuel is the primary fuel type (>98%) in the baseline. Also, the woodfuel using institutions were found to be using traditional /three stone fire stove for water boiling and the charcoal using institutions were found to be using traditional coal pot

Types /Efficiency of water boiling systems:

The following table presents the type of baseline devices being used for water boiling in schools/institutions. The average thermal efficiency has been considered as follows as conservative measure:

Baseline thermal efficiency	
Type of baseline stove	% Value
Traditional/three-stone fire woodfuel	10%
Traditional Charcoal pot	15%
Kerosene stove	50%
Baseline thermal efficiency	12.53%

The weightage average thermal efficiency of the baseline devices is therefore being considered as 12.53% (refer ER calculator for details).

The project WPS will result in elimination/reduction of non-renewable biomass / fossil fuels usage for water boiling. This will result in significant improvement in indoor air pollution associated with use of solid biomass / fossil fuel based inefficient cooking and water boiling practice which has a direct correlation with respiratory illness and mortality rates, especially among women and children, worldwide.

Thus, the baseline scenario as per 3.4.1 and 3.4.2 (Suppressed demand) of the methodology is the use of non-renewable biomass/fossil fuels for water boiling. Since the voluntary Project Activity (VPA) is accounting Suppressed Demand Scenario to establish baseline, hence, the scale of VPA is limited to small scale. As per para 1.1.1(c) of the GS4GG Rule Update "Application of Suppressed Demand, project type and applicable scale threshold (entry into force: 13/11/2020)" project activity shall not result in GHG emission reductions more than 60,000 tCO₂e per year in any year of the crediting period. Refer section B.6.4 below where the estimated ERs from the VPA are projected to be lower than 60,000 tCO₂e per annum over the crediting period. In case, if any year during the crediting period, the annual ERs from the VPA exceeds 60,000 tCO₂e, they shall be capped at 60,000 tCO₂e.

B.5. Demonstration of additionality

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Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Activity Requirement: As per GS4GG Community services activity requirements, Version 1.2, Para 4.1.9, Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of design certification: (a) Positive list (Annex B of this document) (b) Projects located in LDC, SIDS, LLDC (c) Microscale projects Justification:
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	As per Positive List - Annex B, the project meets the criteria given in 1.1.3 <i>Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year.</i> The project meets the requirements of Positive List mentioned within the CSA Requirements and hence is deemed additional.
Describe how the proposed project meets the criteria for deemed additionality.	Each project unit shall result in emission reductions less than 600 tCO ₂ e per annum. In case the VPA generates more than 60,000 tCO ₂ e per annum, the VERs shall be capped at 60,000 tCO ₂ e per annum

B.5.1. Prior Consideration

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Start date of VPA is 12/01/2018 (date of installation of first unit in the VPA) and the VPA crediting period start date under GS is 01/01/2021. The systems in the VPA have transitioned from CDM PoA 9948, hence, prior consideration is not applicable.

B.5.2. Ongoing Financial Need

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Not Applicable. The project is automatically additional as per Annex B of CSA Requirements as explained in B.5 above.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs:

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
1 No Poverty	1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance	1.4.1 Proportion of population living in households with access to basic services Indicator: Total number of premises (Schools / institutions) with at least one WPS distributed / installed under the project

3 Good Health and Well Being	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	3.9.2 Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and Hygiene for All (WASH) services) Indicator: % sampled Schools / institutions reporting reduction in incidence of diarrhoea and water borne diseases etc. after shifting to the project WPS
6 Clean Water and sanitation	6.1 By 2030, achieve universal and equitable access to safe and affordable drinking water for all	6.1.1 Proportion of population using safely managed drinking water services Indicator: % of WPS distributed/installed providing safe drinking water quality
7 Affordable and Clean Energy	7.1 By 2030, ensure universal access to affordable, reliable, and modern energy services	7.1.2 Proportion of population with primary reliance on clean fuels and technology Indicator: % Schools / institutions reporting an operational WPS in project
8 Decent Work and Economic Growth	8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities Indicator: Number of male / females employment created by project
13 Climate Action (mandatory)	Not applicable	Amount of CO ₂ e emissions reduced by the project per year.

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

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Technologies employed under the VPA include low-carbon water purification technologies intended as technologies and/or practices under the methodology. The VPA applies the following methodological choices:

As the expected technical life of the project technology is shorter than the crediting period, end users will be provided replacement systems of comparable quality in order to continue claiming emission reductions or otherwise, the ERs will be limited to the expected technical life.

$f_{NRB,f,y}$ is being fixed ex-ante as option provided in parameter table SDWS 21 of the methodology.

The following default values for water consumption (used for drinking, food preparation and cleaning) as per the meth will be used for ER calculations:

Type of Premises	Default value	Applicability	Reference
Full-day premises	4 L /person / day	Premises like households etc.	Emission reductions from Safe Drinking Water Supply Version 1.0 – 03/5/2021”
Boarding school	4 L /person / day	-	
Half time premises	3 L /person / day	Premises like day schools, offices etc.	

The VPA applies the following equations:

SDG 1: No Poverty

Applied methodology/approach	Equation/calculation
1.4.1 Proportion of population living in households with access to basic services Approach: Total number of premises with at least one WPS distributed / installed under the project	1. WPS distribution/installation records Net Benefit (SDG 1) = $ABS_{Project} - ABS_{Baseline}$ Where: $ABS_{Baseline}$ Access to basic services (number of premises with at least one WPS distributed / installed under the project in baseline) $ABS_{Project}$ Access to basic services (number of premises with at least one WPS distributed / installed under the project in Project)

SDG 3: Good Health and Well Being

Applied methodology/approach	Equation/calculation
3.9.2 Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and Hygiene for All (WASH) services Approach: % of users reporting reduction in incidence of diarrhoea and water borne diseases etc. after shifting to the project WPS	Ex-post Monitoring Surveys Records Net Benefit (SDG 3) = $IH_{USERS,Project} - IH_{Baseline}$ Where: $IH_{Baseline}$ % of users reporting reduction in incidence of diarrhoea and water borne diseases etc. in baseline $IH_{Project}$ % of users reporting reduction in incidence of diarrhoea and water borne diseases etc. after shifting to the project WPS

SDG 6: Clean water and Sanitation

Applied methodology/approach	Equation/calculation
6.1.1 Proportion of population using safely managed drinking water services Approach: 1. % of WPS distributed/installed provide safe drinking water quality	1. Ex-post Monitoring Surveys Records $\text{Net Benefit (SDG 6)} = \text{SWQ}_{\text{Project}} - \text{SWQ}_{\text{Baseline}}$ Where: $\text{SWQ}_{\text{Baseline}}$ % users reporting safe water quality in baseline $\text{SWQ}_{\text{Project}}$ % users reporting safe water quality in baseline

SDG 7: Affordable and Clean Energy

Applied methodology/approach	Equation/calculation
7.1.2 Proportion of population with primary reliance on clean fuels and technology Approach: Monitor the % of operational WPS distributed	WPS distribution records + Ex-post Monitoring Surveys Records $\text{Net Benefit (SDG 7)} = \text{AAC}_{\text{Project}} - \text{AAC}_{\text{Baseline}}$ Where: $\text{AAC}_{\text{Baseline}}$ Access to affordable and clean energy (% of operating WPS units under Baseline) $\text{AAC}_{\text{Project}}$ Access to affordable and clean energy (% of operating WPS units under Project)

SDG 8: Decent Work and Economic Growth

Applied methodology/approach	Equation/calculation
8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities Approach: Recording the number of employees (male / female) in the project under administrative, sales, production and management positions	Employment records $\text{Net Benefit (SDG 8)} = \text{QE IG}_{\text{Project}} - \text{QE IG}_{\text{Baseline}}$ Where: $\text{QE IG}_{\text{Baseline}}$ Quantitative Employment and income generation (Number of person (male and female) hired under Baseline) $\text{QE IG}_{\text{Project}}$ Quantitative Employment and income generation (Number of person (male and female) hired under Project)

SDG 13: Climate Action

Applied methodology/approach	Equation/calculation
Amount of CO ₂ e emissions reduced by the project per year	Water Purification Systems For water purification technologies the overall GHG reductions achieved by the

<u>Water Purification Systems</u> Approach: Emission reductions from Safe Drinking Water Supply Version 1.0 – 03/5/2021	project activity in year y are calculated as follows: The baseline emission factor shall be calculated as follows: $EF_b = SE_{w,b,y} * \sum (x_f * (EF_{b,f,CO2} * f_{NRB,f,y} + EF_{b,f,nonCO2})) \div 10^9$ Where: EF_b = Emission factor for the use of fuel to obtain safe water in the baseline (tCO2e/L) $SE_{w,b,y}$ = Specific energy required to boil water (kJ/L), to be calculated as per the paragraph below x_f = Proportion of fuel f used in the baseline (fraction determined based on an energy basis) $EF_{b,f,CO2}$ = CO2 emission factor from use of fuel f (tCO2/TJ) $EF_{b,f,nonCO2}$ = Non-CO2 emission factor arising from use of fuel f, when the baseline fuel f is biomass or charcoal (tCO2e/TJ). This parameter is omitted when f is a fossil fuel. $f_{NRB,f,y}$ = Fractional non-renewability status of woody biomass fuel during year y (fraction). For biomass, it is the fraction of woody biomass that can be established as non-renewable. This parameter is omitted when f is a fossil fuel. f = Index for baseline fuel types . The specific energy required to boil water using the baseline technology ($SE_{w,b,y}$) is determined as follows, by calculating the energy input required to obtain 1 L of boiling water, including boiling and vaporization losses²⁰, taking
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²⁰ The previous version of TPDDTEC Annex 3 assumed that purifying water by boiling would require boiling water for 10 minutes. This assumption is revised to 5 minutes, following WHO technical information that less than 5 minutes of boiling is sufficient for inactivation of enteric bacteria (Technical Brief WHO/FWC/WSH/15.02, 2015).

	into account default or measured stove efficiency. $SE_{w,b,y} = 360.83/\eta_{wb}$ Where: 360.83 = Default amount of energy required to obtain 1 L of water after 5 minutes of boiling from a first principles approach kJ/l η_{wb} = Efficiency of the stoves for baseline water boiling (%). Weighted average of baseline stove types. The baseline emissions shall be calculated as follows: $BE_y = EF_b \times (1 - C_b - X_{cleanboil,y}) \times Q_y \times M_{q,y}$ Where: BE_y = Baseline emissions from the use of fuel to obtain safe water in the baseline (tCO₂e) C_b = Proportion of project end-users who in the baseline were already using a safe water supply that did not require boiling (%) $X_{cleanboil,y}$ = Proportion of project end-users that boil safe water in the project year y (%) Q_y = Quantity of safe drinking water provided by the project in year y (L) $M_{q,y}$ = Modifier for the water quality in year y In the case of IWT, the quantity of safe drinking water provided by the project Q_y is determined as follows: $Q_y = \sum N_{p,y} \times U_{p,y} \times QPW_{hh,p,y} \times DP_{p,y}$ Where: $N_{p,y}$ = Number of premises type p with at least one project technology in year y
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	$U_{p,y}$ = Usage rate of the project technology by premises type p during year y (%) $QPW_{hh,p,y}$ = Volume of drinking water per premises p per day in year y (L) $DP_{p,y}$ = Days the project technology is present for end-users in the premises p in year y The volume of drinking water per premises is determined by considering whether the capacity of the project device is sufficient to provide at least the default amount of drinking water, as follows: $QPW_{hh,p,y} = \min((q_i \times DN_{p,y}), (QPW_p \times HN_{p,y}))^{21}$ Where: q_i = Capacity of the HWT or IWT individual project technology supplied (L) $DN_{p,y}$ = Average number of individual project technologies in each project premises type p in year y $HN_{p,y}$ = Number of individuals per premises type p (e.g. household, school) in year y QPW_p = Volume of drinking water per person per day for premises type p (L) * average number of operational days in the year. Apply the default value or monitored value through water consumption field tests in the project scenario, capped at 5.5 L per person per day
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²¹ The ER calculation approach will ensure that the credits are claimed only for the period for which the school / institution is operational and has treatment supplies available. As a conservative measure, if there is a supply shortfall in a school / institution, the VERs calculated are limited by the available treatment capacity i.e. VERs will be claimed for minimum of the Available treatment capacity Vs total quantity of treated water required. This ensures that any period affected by shortfall / dis-continuity in the supplies is not accounted in the emission reductions.

	Project emissions are not envisaged in the VPA as chlorination does not require any fuel/ electricity use. The emission reductions are calculated as follows: $ER_y = BE_y - PE_y - LE_y$ Where: ER_y = Emission reductions in year y (t CO₂e/yr) BE_y = Baseline emissions in year y (t CO₂e/yr) PE_y = Project emissions in year y (t CO₂e/yr) LE_y = Leakage emissions in year y (t CO₂e/yr)
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Leakage

In line with the methodology "Emission reductions from Safe Drinking Water Supply" Version 1.0 – 03/5/2021, the following leakage studies will be included to assess the emission reductions generated by the activity:

1. The following potential sources of leakage have been evaluated ex-ante:

a) Members of the population who do not participate in the project, and previously used lower emitting energy sources, instead use the nonrenewable biomass saved under the project activity

- There is no evidence that the project increases the use of higher emitting fuel outside the project boundary where lower emitting energy sources take in place. Woody biomass fuels remain a valuable and declining resource. Moreover, many users collect locally available wood for free. The free supply of wood is external to economic markets for wood and thus, to a large extent, these communities are shielded from market forces on wood fuel prices. This trend will not be reversed by the project activity.
- The government published data provides evidence that both non-project and project beneficiaries mainly use firewood and/or charcoal to fuel traditional cookstoves for water boiling. Baseline surveys also substantiate that solid biomass fuels is the primary fuel type (>98%) in the baseline. Also, the woodfuel using institutions were found to be using traditional /three stone fire stove for water boiling and the charcoal using institutions were found to be using traditional coal pot. There is therefore no risk that the non-project beneficiaries may adopt higher emitting energy sources due to the project activity.

- Given that the leakage assessment does not expect an increase in fuel consumption by the non-project beneficiaries attributable to the project activity, calculations do not need to be adjusted to account for this leakage at the time of validation. However, the relevant parameters/data may be updated, if necessary, as specified by the monitoring plan.

Overall justification at the time of validation: No leakage is present

b) The project significantly reduces the NRB fraction within an area where other GHG mitigation project activities account for NRB fraction in their baseline scenario.

The impact of the project activity on the NRB fraction within the project boundary is expected to be negligible. The size of the project activity is small compared to the scale of the project boundary (country of Nigeria) and thus is not likely to be significant within the target population to affect a change in fNRB. The amount of woody biomass saved due to project activity is very less as compared to the total woody biomass consumption in Nigeria i.e., 229,692,183 Tonnes/yr (refer C4 Ecolution report for NRB fraction calculation). Hence It will not have any significant impact on the NRB fraction of Nigeria.

Overall justification at the time of validation: No leakage is present

c) The project population compensates for loss of the space heating effect of water boiling by adopting some other form of space heating or by retaining some baseline wood fuel-burning practices.

- The climate throughout Nigeria is temperate to hot. Space heating is not likely to be significant within the target population. No evidence exists that the project will result in increased fuel use for heating from inefficient stoves.

Overall justification at the time of validation: No leakage is present

2. Second, for each source for which the leakage assessment expects an increase in non-renewable biomass fuel consumption by non-project households/users attributable to the project activity, then calculations must be undertaken to account for the leakage from this source. Leakage is either calculated as a quantitative emissions volume (tCO₂e) or as a percentage of total emission reductions. The project documentation shall include a projection of leakage emissions based on available data and information. The monitoring plan must include monitoring parameters to be registered during the leakage investigation every two years to populate the leakage calculation.

Justification: Not Applicable as no significant increase in non-renewable biomass fuel consumption by non-project households/users attributable to the project activity is envisaged

3. Third, the project developer must conduct a leakage investigation every two years using relevant methods. For example, surveys to determine parameters for the leakage calculation may be combined with project monitoring surveys, as is applicable.

Justification: leakage will be determined every two years.

4. If the ex-ante evaluation shows that leakage emissions are less than 5% of total emission reductions, then no monitoring is needed, and emission reductions simply shall be adjusted 5% down. In this case, the sources and magnitude of leakage emissions must be reassessed at the time of crediting period renewal.

Justification: As per 3 above, the PD will estimate leakage every two years. Alternatively, in case the leakage cannot be assessed, due to lack of data or other reasons, the PD may use a leakage discount factor of 0.95 (given it has been demonstrated above already that leakage is 0).

Ex-ante emission reduction calculations are provided in the Emission Reduction Calculation spreadsheet, submitted to DOE separately. In addition, further information for each parameter is provided in sections B.6.2 and B.7.1 of this VPA-DD.

B.6.2. Data and parameters fixed ex ante

SDG Indicator	SDG 1: No Poverty
Data/parameter	ABS _{Baseline}
Unit	Number
Description	Access to Basic Services (number of premises with at least one WPS distributed / installed under the project under the baseline)
Source of data	--
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 1 Impact calculation
Additional comment	--

SDG Indicator	SDG 3: Good Health and Well Being
Data/parameter	IH _{Baseline}
Unit	%
Description	% of users reporting reduce in incidence of diarrhoea and water borne diseases etc. in baseline (improved health)
Source of data	--
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 3 Impact calculation
Additional comment	--

SDG Indicator	SDG 6: Clean Water and sanitation
Data/parameter	SWQ _{baseline}
Unit	%
Description	% users reporting safe water quality in baseline
Source of data	--
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 6 Impact calculation
Additional comment	--

SDG Indicator	SDG 7: Affordable and Clean Energy
Data/parameter	AAC _{Baseline}
Unit	%
Description	Access to affordable and clean energy (% of operating WPS units under Baseline)
Source of data	--
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 7 Impact calculation
Additional comment	--

SDG Indicator	SDG 8: Decent Work and Economic Growth
Data/parameter	QE IG _{Baseline}
Unit	number
Description	Quantitative Employment and income generation (Number of person (male and female) hired under Baseline)
Source of data	--
Value(s) applied	0
Choice of data or Measurement methods and procedures	--
Purpose of data	SDG 8 Impact calculation
Additional comment	--

SDG 13 parameters are governed by the methodology. The following parameters have not been included as these are deemed not applicable to the VPA:

1. Parameters applicable to CWT / CWS:
 - a. Number of household/institution per CWT/CWS
 - b. Project technology performance level (CWT or CWS)
2. NCV_f – fossil fuel consumption in project scenario is not applicable
3. EF_f – fossil fuel consumption in project scenario is not applicable

SDG Indicator	SDG 13: Climate Change																	
Data/parameter	Project technology description																	
Unit	NA																	
Description	The detailed description of the planned project technology																	
Source of data	- Manufacturer specifications - Third-party certification by a qualified entity, for example recognized certification agency by National/International Standard body																	
Value(s) applied	<table><tr><td>Description</td><td>UltraFLO</td><td>UltraTAB</td></tr><tr><td>Manufacturer</td><td>Medentech</td><td>Medentech</td></tr><tr><td>Product Name</td><td>UltraFLO</td><td>Big Pack, Small Pack</td></tr><tr><td>Technology type</td><td>Chlorination</td><td>Chlorination</td></tr><tr><td>Performance classification</td><td>Complies with National standard</td><td>Complies with National standard</td></tr></table>			Description	UltraFLO	UltraTAB	Manufacturer	Medentech	Medentech	Product Name	UltraFLO	Big Pack, Small Pack	Technology type	Chlorination	Chlorination	Performance classification	Complies with National standard	Complies with National standard
Description	UltraFLO	UltraTAB																
Manufacturer	Medentech	Medentech																
Product Name	UltraFLO	Big Pack, Small Pack																
Technology type	Chlorination	Chlorination																
Performance classification	Complies with National standard	Complies with National standard																
Choice of data or Measurement methods and procedures	As per methodology																	
Purpose of data	--																	
Additional comment	--																	

SDG Indicator	SDG 13: Climate Change		
Data/parameter	Regulatory framework for safe water supply		
Unit	NA		
Description	National, sub-national and local regulations or guidance for safe drinking water supply, operation and maintenance, including any tariff requirements in host country Nigeria.		
Source of data	National, sub-national and local authorities		
Value(s) applied	President of Nigeria launched the National Action Plan (NAP), a 13-year strategy for the Revitalization of Nigeria's Water Supply, Sanitation, and Hygiene (WASH) Sector . The goal of the Revitalization Strategy is to ensure that all Nigerians have access to sustainable and safely managed WASH services by 2030, in compliance with the Sustainable Development Goals (SDG) for Water (Goal 6.1) and Sanitation (Goal 6.2).		

	National Water Resources Bill- 2020, enacted by National assembly of Federal Republic of Nigeria, establishes regulatory framework for the water resource sector in Nigeria and provide framework for the equitable and sustainable development, management, use and conservation of Nigeria's surface and ground water resources. The objective of the bill is to also ensure the citizen's right of access to safe water and basic sanitation. The establishment of Nigerian Standard for Drinking Water Quality (NSDQW) ensures the safety of the drinking water supplies and the protection of public health. The implementation of the project is fully in line with the relevant water resources policies/framework of Nigeria. The project does not undermine or conflict with any national, sub-national and local regulations or guidance for safe drinking water supply, operation, and maintenance, including any tariff requirements. Further, the national standards and local regulations for safe drinking water supply do not impose any cap on parameters used by the methodology and therefore have no implications on emission reduction calculations.
Choice of data or Measurement methods and procedures	--
Purpose of data	--
Additional comment	Update at the start of crediting period

SDG Indicator	SDG 13: Climate Change
Data/parameter	Water sources in the project boundary
Unit	NA
Description	Improved and Unimproved drinking water sources in Nigeria
Source of data	Nigeria Schools and Institutions: Baseline survey

Value(s) applied	<table><tr><th>S.No</th><th>Drinking water source</th><th>% Premises</th></tr><tr><td>1.</td><td>Government/ Private piped connection</td><td>3.53</td></tr><tr><td>2.</td><td>Surface water</td><td>2.35</td></tr><tr><td>3.</td><td>Well, /Borehole</td><td>94.12</td></tr></table>			S.No	Drinking water source	% Premises	1.	Government/ Private piped connection	3.53	2.	Surface water	2.35	3.	Well, /Borehole	94.12
	S.No	Drinking water source	% Premises												
	1.	Government/ Private piped connection	3.53												
	2.	Surface water	2.35												
	3.	Well, /Borehole	94.12												
	As per Annex 2 of the applied GS methodology, piped water and water from boreholes or protected wells belong to improved sources of drinking water. The rest belong to unimproved sources of drinking water as follows:														
<table><tr><th>S.No.</th><th>Source of drinking water</th><th>% Premises</th></tr><tr><td>1.</td><td>Improved sources</td><td>97.65%</td></tr><tr><td>2.</td><td>Unimproved Sources</td><td>2.35%</td></tr></table>			S.No.	Source of drinking water	% Premises	1.	Improved sources	97.65%	2.	Unimproved Sources	2.35%				
S.No.	Source of drinking water	% Premises													
1.	Improved sources	97.65%													
2.	Unimproved Sources	2.35%													
Choice of data or Measurement methods and procedures	--														
Purpose of data	--														
Additional comment	Update at the start of crediting period														

SDG Indicator	SDG 13: Climate Change														
Data/parameter	Stove technologies used in the project boundary														
Unit	NA														
Description	The proportion of different stove types used in premises in the geographical area of the project. If the project covers different types of end-users premises (e.g. households, institutions), then the stoves technologies should be determined for each premises type.														
Source of data	Nigeria Schools and Institutions: Baseline survey														
Value(s) applied	<table><tr><th>S.no</th><th>Stove Technology</th><th>School/Institutions (%)</th></tr><tr><td>1</td><td>Traditional / 3 Stone Fire</td><td>58.02</td></tr><tr><td>2</td><td>Traditional Charcoal Pot</td><td>40.74</td></tr><tr><td>3</td><td>Kerosene Stove</td><td>1.23</td></tr></table>			S.no	Stove Technology	School/Institutions (%)	1	Traditional / 3 Stone Fire	58.02	2	Traditional Charcoal Pot	40.74	3	Kerosene Stove	1.23
S.no	Stove Technology	School/Institutions (%)													
1	Traditional / 3 Stone Fire	58.02													
2	Traditional Charcoal Pot	40.74													
3	Kerosene Stove	1.23													
Choice of data or Measurement methods and procedures	As per methodology														
Purpose of data	Determination of baseline emissions														
Additional comment	Update at the start of crediting period														

SDG Indicator	SDG 13: Climate Change														
Data/parameter	Expected technical life of project technology														
Unit	Treatment volume or operational hours or time period (e.g., “eight years”)														
Description	The expected technical life of an individual project technology shall be defined in the PDD. The details include both technology/device life and filter life, if a filter is used and it is replaceable.														
Source of data	Manufacturer specifications														
Value(s) applied	<table><tr><th>Description</th><th>UltraFLO</th><th>UltraTAB</th></tr><tr><td>Treatment Volume (Ltrs)</td><td>340,000</td><td>Big Pack: 48,000 Small Pack: 10,000</td></tr><tr><td>Life Span / Expiry - filter</td><td>5 years</td><td>5 years</td></tr><tr><td>Life Span / Expiry - system</td><td>30 years</td><td>30 years</td></tr></table>			Description	UltraFLO	UltraTAB	Treatment Volume (Ltrs)	340,000	Big Pack: 48,000 Small Pack: 10,000	Life Span / Expiry - filter	5 years	5 years	Life Span / Expiry - system	30 years	30 years
Description	UltraFLO	UltraTAB													
Treatment Volume (Ltrs)	340,000	Big Pack: 48,000 Small Pack: 10,000													
Life Span / Expiry - filter	5 years	5 years													
Life Span / Expiry - system	30 years	30 years													
Choice of data or Measurement methods and procedures	Manufacturer specifications														
Purpose of data	Determination of baseline emissions														
Additional comment	The lifetime of the systems is not constrained. Instead, the system lifetime is governed by the lifetime of the cartridge / Tab Pack as applicable, depending on the type of system installed/distributed. In case of UltraFLO/UltraTAB, the expiry of the cartridge/TABPack is 5 years. Every-time a school receives a new supply of UltraFLO cartridge/TABPack, the lifetime of the system is automatically deemed renewed, the supplies being a consumable. Thus, the systems are not constrained with respect to lifetime and can extend much beyond the crediting period. Therefore “Life Span / Expiry – system” covers the total length of crediting period of the VPA. In case the requirement for replacement due to loss or functional issues arises then a replacement system of comparable or better technology will be installed/distributed and relevant documents will be provided to end users														

SDG Indicator	SDG 13: Climate Change		
Data/parameter	X_f		
Unit	Percentage of fuel f use in target population		

Description	The proportion of each different cooking fuel f used in the project boundary by end-users:														
Source of data	Nigeria Schools and Institutions: Baseline Survey														
Value(s) applied	<table><tr><th>S.no</th><th>Baseline Fuel</th><th>School/Institutions (%)</th></tr><tr><td>1.</td><td>Woodfuel</td><td>58.02</td></tr><tr><td>2.</td><td>Charcoal</td><td>40.74</td></tr><tr><td>3.</td><td>Kerosene</td><td>1.23</td></tr></table>	S.no	Baseline Fuel	School/Institutions (%)	1.	Woodfuel	58.02	2.	Charcoal	40.74	3.	Kerosene	1.23		
S.no	Baseline Fuel	School/Institutions (%)													
1.	Woodfuel	58.02													
2.	Charcoal	40.74													
3.	Kerosene	1.23													
Choice of data or Measurement methods and procedures	As per methodology baseline surveys were conducted														
Purpose of data	Determination of baseline emissions														
Additional comment	Update at the start of crediting period As per the baseline study conducted, ~98% of the sampled schools relying on solid biomass fuel (~58% wood, ~40% charcoal). The value obtained has been cross checked against 2016 report available from Global Alliance on clean cooking i.e., "THE TRUTH ABOUT COOKING LANDSCAPE ANALYSIS" that states the following as a general statement: "90 million Nigerians, and almost all public institutions, cook with wood on the traditional "three-stone fire." Thus, the GACC report indicates ~100% public institutions relying on solid biomass fuel, which supports the result obtained from the baseline surveys conducted.														

SDG Indicator	SDG 13: Climate Change
Data/parameter	EF_{b,f,CO_2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor from use of fuels
Source of data	IPCC defaults
Value(s) applied	Wood = 112 tCO ₂ /TJ Charcoal = 165.22 tCO ₂ /TJ (includes charcoal production emissions)
Choice of data or Measurement methods and procedures	As per applied methodology
Purpose of data	Determination of baseline emissions
Additional comment	--

SDG Indicator	SDG 13: Climate Change
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Data/parameter	$EF_{b,f,nonCO_2}$
Unit	tCO _{2e} /TJ
Description	Non-CO ₂ emission factor from use of fuels, in case the baseline fuel is biomass or charcoal
Source of data	IPCC defaults
Value(s) applied	AR5 GWP - Wood: 9.46 tCO _{2e} /TJ - Charcoal: 44.83 tCO _{2e} /TJ (includes production emissions of CH ₄ and N ₂ O)
Choice of data or Measurement methods and procedures	Deemed valid by GS VER Methodology
Purpose of data	Determination of baseline emissions
Additional comment	--

SDG Indicator	SDG 13: Climate Change		
Data/parameter	η_{wb}		
Unit	Percentage		
Description	Weighted average efficiency of the baseline water boiling devices.		
Source of data	Nigeria Schools and Institutions: Baseline Survey		
Value(s) applied	Methodology default value / Credible published literature		
	Nigeria-Schools and Institutions		
	Stove Technology:		
	S.no	Stove Technology	School/Institutions (%)
	1	Traditional / 3 Stone Fire	58.02
	2	Traditional Charcoal Pot	40.74
3	Kerosene Stove	1.23	
	S.no	Stove Technology	% Efficiency
	1	Traditional / 3 Stone Fire	10
	2	Traditional Charcoal Pot	15
	3	Kerosene Stove	50
	Thus, $\eta_{wb} = (58.02\% * 0.1) + (40.74\% * 0.15) + (1.23\% * 0.5) = 12.53\%$		
	Choice of data or Measurement methods and procedures	As per the methodology	
Purpose of data	Determination of baseline emissions		

Additional comment	--
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SDG Indicator	SDG 13: Climate Change												
Data/parameter	C_b												
Unit	Percentage												
Description	Proportion of project end-users who in the baseline were already using safe water, either from an improved water source, or from a water treatment method other than boiling.												
Source of data	Baseline Survey												
Value(s) applied	<table border="1"> <thead> <tr> <th colspan="2">Baseline Treatment Method</th></tr> <tr> <th>Type of treatment method</th><th># of Schools</th></tr> </thead> <tbody> <tr> <td>Receiving safe water without treatment</td><td>2</td></tr> <tr> <td>treating/ would have treated water without boiling</td><td>6</td></tr> <tr> <td>treating/ would have treated water via boiling</td><td>162</td></tr> <tr> <td>C_b</td><td>4.71%</td></tr> </tbody> </table>	Baseline Treatment Method		Type of treatment method	# of Schools	Receiving safe water without treatment	2	treating/ would have treated water without boiling	6	treating/ would have treated water via boiling	162	C_b	4.71%
Baseline Treatment Method													
Type of treatment method	# of Schools												
Receiving safe water without treatment	2												
treating/ would have treated water without boiling	6												
treating/ would have treated water via boiling	162												
C_b	4.71%												
Choice of data or Measurement methods and procedures	Value is calculated based on baseline survey conducted for the target users												
Purpose of data	Determination of baseline emissions												
Additional comment	Cross check measures: Study referenced for cross check in GS decision for deviation DEV_259– Demographic and Health Survey (DHS), Nigeria 2019 The DHS report only provides information for households and does not include any information on schools in Nigeria. Thus, DHS study does not provide for an appropriate cross-check reference. Even if this is considered, the % of households using treatment methods other than boiling is merely ~2% as per DHS, table 2.1.3, page 23. Thus, as per cross checks conducted, the C_b value established as 4.71% is conservative.												

SDG Indicator	SDG 13: Climate Change
Data/parameter	q_i
Unit	Liters/cartridge or Liters/Tab Pack

Description	Capacity of the household or institutional water treatment technology		
Source of data	Manufacturer specifications		
Value(s) applied	Description	UltraFLO	UltraTAB
	Dosage rate (Treatment Capacity)	340,000L/Cartridge	Big Pack: 48,000L/Tab Small Pack: 10,000L/Tab
Choice of data or Measurement methods and procedures	Manufacturer specifications		
Purpose of data	Determination of baseline emissions		
Additional comment	An alternative approach is being used given the project technologies does not have a standard flow rate.		

SDG Indicator	SDG 13: Climate Change
Data/parameter	$f_{nr,b,f,y}$
Unit	Percentage
Description	Fractional non-renewability status of woody biomass fuel during year y , in case the baseline fuel is biomass or charcoal
Source of data	C4 Ecolutions Report
Value(s) applied	96
Choice of data or Measurement methods and procedures	Calculated using Tool 30 of CDM
Purpose of data	Determination of baseline emissions
Additional comment	-

SDG Indicator	SDG 13: Climate Change
Data/parameter	QPW_p
Unit	Litres per person per day
Description	Volume of drinking water per person per day for premises type p
Source of data	Parameter table SWDS 24 of methodology

Value(s) applied	Type of Premises	Default value	Applicability
	Full-day premises	4 L /person / day	Premises like households etc.
	Boarding school	4 L /person / day	-
	Half time premises	3 L /person / day	Premises like day schools, offices etc.
Choice of data or Measurement methods and procedures	Default from methodology		
Purpose of data	Baseline emission calculations		
Additional comment	-		

B.6.3. Ex ante estimation of SDG Impact

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Sustainable Development Goals Targeted	SDG Impact Indicator	Calculation	Ex-ante Estimate (Annual)
1 No Poverty	Indicator: number of premises with at least one WPS distributed / installed under the project	$ABS_{Project} - ABS_{Baseline}$	650
3 Good Health and Well Being	Indicator: % of users reporting reduce in incidence of diarrhoea and water borne diseases etc. (improved health)	$IH_{Project} - IH_{Baseline}$	90%
6 Clean Water and sanitation	Indicator: % WPS distributed / installed providing safe drinking water to beneficiaries	$SWQ_{Project} - SWQ_{Baseline}$	90%
7 Affordable and Clean Energy	Indicator: % users reporting an operational WPS in project	$AAC_{Project} - AAC_{Baseline}$	90%
8 Decent Work and Economic Growth	Indicator: Number of male / female numbers of employment created by project	$QE\ IG_{Project} - QE\ IG_{Baseline}$	19
13 Climate Action	Amount of CO ₂ e emissions reduced by the project per year	Refer the table below	59,312 tCO ₂ e

Water Purification System (WPS)				
Description	Symbol	Reference	Unit	VPA 01
Percentage of fuel f use in target population	xf	woodfuel users	Percentage	58.02%
Percentage of fuel f use in target population	xf	Charcoal users	Percentage	40.74%
Percentage of fuel f use in target population	xf	Kerosene users	Percentage	1.23%
CO ₂ Emission factor for Wood	EF _{b,f,CO2} / EF _{p,f,CO2}	Meth default value, Fixed Ex-ante	tCO ₂ /TJ	112.0
CO ₂ Emission factor for charcoal	EF _{b,f,CO2} / EF _{p,f,CO2}	Meth default value, Fixed Ex-ante	tCO ₂ /TJ	165.2
CO ₂ Emission factor for kerosene	EF _{b,f,CO2} / EF _{p,f,CO2}	IPCC 2006 GHG Inventories, Volume 2, Table 1.4, Fixed Ex-ante	tCO ₂ /TJ	71.9
Non-CO ₂ emission factor for Wood	EF _{b,f,nonCO2} / EF _{p,f,nonCO2}	Meth default value, Fixed Ex-ante	tCO ₂ e/TJ	9.46
Non-CO ₂ emission factor for charcoal	EF _{b,f,nonCO2} / EF _{p,f,nonCO2}	Meth default value, Fixed Ex-ante	tCO ₂ e/TJ	44.83
Non-CO ₂ emission factor for kerosene	EF _{b,f,nonCO2} / EF _{p,f,nonCO2}	IPCC 2006 GHG Inventories, Volume 2, Table 2.5, Fixed Ex-ante	tCO ₂ e/TJ	0.439
Average efficiency of the baseline water boiling devices	nwb	for Traditional 3-stone fire, Baseline survey	Percentage	12.5%
Specific energy required to boil 1 L of water using the baseline technology	Sew,b,y	Calculated	kJ/L	2879.53
Fractional non-renewability status of woody biomass fuel during year y, in case the baseline fuel is biomass or charcoal	f _{NRB,f,y}	Calculated as per Tool 30 using C4 Ecolutions Report	Percentage	96.0%
Emission factor for use of fuel to obtain safe water in the baseline	EF _b	calculated	tCO ₂ /L	0.00043659
Portion of users of project technology who were already in baseline consuming safe water without boiling it	Cb	Baseline Survey, fixed ex-ante	Percentage	4.71%
Proportion of project end-users that boil safe (treated, or from safe supply) water after installation of project technology in year y	x _{cleanboil,y}	Assumed for ex-ante calculation only, usage survey will be conducted for ex post value	Percentage	0.00%
Water Quality	Mq,y	Assumed for ex-ante calculation only, WQT will be conducted for ex post value	fraction	0.90
Volume of drinking water per person per day for full day premises	QPWp,full day	Meth default value, Fixed Ex-ante	Liters/pers on/day	4.00
Volume of drinking water per person per day for half day premises	QPWp, half day	Meth default value, Fixed Ex-ante	Liters/pers on/day	3.00
% full day premises	--	Assumed	%	7%
Volume of drinking water per person per day for premises type p	QPWp	calculated	Liters/pers on/day	3.07
Number of individuals per premises type p in the project boundary in year	HNp,y	Assumed for ex-ante calculation only, project usage survey will be conducted for ex post value	Persons	420
Usage rate of the project technology by premises type p during year y	U _{p,y}	Assumed for ex-ante calculation only, in person or telephonic survey will be conducted for ex post value	Percentage	90.0%
Average days the project technology is present for end-users in the premises p in year y	DPp,y	Assumed for ex-ante calculation only. Sales/Distribution database will be used for ex post determination	days	210
Accumulated number of premises type p with at least one individual project technology in year y	Np,y	Calculated	Number	650
Baseline Emissions	BEy	Calculated	tCO₂	59,312.00
Project Emissions				
Project Emissions	PEy	Calculated	tCO₂	-
Leakage Emissions				
Leakage emissions	LEy	not applicable	tCO₂	-
Emission Reduction				
Emission Reductions	ERy	Calculated	tCO₂	59312

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 1: No Poverty

Year	Baseline estimate (ABS _{Baseline})	Project estimate ABS _{Project})	Net benefit
2021	0	650	650
2022	0	650	650
2023	0	650	650

2024	0	650	650
2025	0	650	650
Total	0	650	650
Total number of crediting years	5		
Annual average over the crediting period	0	650	650

SDG 3: Good Health and Well Being

Year	Baseline estimate (IH _{Baseline})	Project estimate (IH _{Project})	Net benefit
2021	0	90%	90%
2022	0	90%	90%
2023	0	90%	90%
2024	0	90%	90%
2025	0	90%	90%
Total	0	90%	90%
Total number of crediting years	5		
Annual average over the crediting period	0	90%	90%

SDG 6: Clean Water and sanitation

Year	Baseline estimate (SWQ _{Baseline})	Project estimate (SWQ _{Project})	Net benefit
2021	0	90%	90%
2022	0	90%	90%
2023	0	90%	90%
2024	0	90%	90%
2025	0	90%	90%
Total	0	90%	90%
Total number of crediting years	5		
Annual average over the crediting period	0	90%	90%

SDG 7: Affordable and Clean Energy

Year	Baseline estimate (AAC _{Baseline})	Project estimate (AAC _{Project})	Net benefit
2021	0	90%	90%
2022	0	90%	90%
2023	0	90%	90%
2024	0	90%	90%
2025	0	90%	90%
Total	0	90%	90%
Total number of crediting years	5		
Annual average over the crediting period	0	90%	90%

SDG 8: Decent Work and Economic Growth

Year	Baseline estimate QE (QE IG _{Baseline})	Project estimate (QE IG _{Project})	Net benefit
2021	0	19	19
2022	0	19	19
2023	0	19	19
2024	0	19	19
2025	0	19	19
Total	0	95	95
Total number of crediting years	5		
Annual average over the crediting period	0	19	19

SDG 13: Climate Action

Year	Baseline estimate	Project estimate	Leakage Estimate	Net benefit
2021	59,312	0	0	59,312
2022	59,312	0	0	59,312
2023	59,312	0	0	59,312
2024	59,312	0	0	59,312
2025	59,312	0	0	59,312
Total	296,560	0	0	296,560

Total number of crediting years	5			
Annual average over the crediting period	59,312	0	0	59,312

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG Indicator	SDG 1: No Poverty
Data / Parameter	ABS _{Project}
Unit	Number
Description	Access to Basic Services (number of premises with at least one WPS distributed / installed under the project under the project)
Source of data	Sales / Installation records
Value(s) applied	650
Measurement methods and procedures	Not applicable
Monitoring frequency	Continuous
QA/QC procedures	-
Purpose of data	SDG 1 contribution
Additional comment	-

SDG Indicator	SDG 3: Good Health and Well Being
Data / Parameter	IH _{Project}
Unit	%
Description	% of users reporting reduce in incidence of diarrhoea and water borne diseases etc. (improved health) in project
Source of data	Ex-post monitoring surveys
Value(s) applied	90
Measurement methods and procedures	Not applicable
Monitoring frequency	Annual / Biennial
QA/QC procedures	-
Purpose of data	SDG 3 contribution
Additional comment	-

SDG Indicator	SDG 6: Clean Water and sanitation
Data / Parameter	SWQ _{Project}

Unit	%
Description	% WPS distributed / installed providing safe drinking water to beneficiaries in project
Source of data	Ex-post water quality tests
Value(s) applied	90
Measurement methods and procedures	Not applicable
Monitoring frequency	Annual / Biennial
QA/QC procedures	-
Purpose of data	SDG 6 contribution
Additional comment	-

SDG Indicator	SDG 7: Affordable and Clean Energy
Data / Parameter	AAC _{Project}
Unit	%
Description	Access to affordable and clean energy (% of operating WPS units under Project)
Source of data	Ex-post monitoring surveys
Value(s) applied	90
Measurement methods and procedures	Not applicable
Monitoring frequency	Annual / Biennial
QA/QC procedures	-
Purpose of data	SDG 7 contribution
Additional comment	-

SDG Indicator	SDG 8: Decent Work and Economic Growth
Data / Parameter	QE IG _{project}
Unit	number
Description	Quantitative Employment and income generation (Number of person (male and female) hired under project)
Source of data	HR records/ Sales and marketing records
Value(s) applied	19
Measurement methods and procedures	Not applicable
Monitoring frequency	Annual / Biennial
QA/QC procedures	-
Purpose of data	SDG 8 contribution
Additional comment	-

SDG 13 parameters are governed by the methodology. The following parameters have not been included as these are deemed not applicable to the VPA:

1. Parameters applicable to CWT / CWS:
 - a. $Q_{m,y}$ - Monitored quantity of safe water provided by the CWS/CWT project in year y
 - b. $HH_{p,y}$ - Number of premises type p served by the project in year y
 - c. $DO_{p,y}$ - Days the project technology is operational for end-users in premises p in year y
2. $f_{NRB,f,y}$ - fixed ex-ante
3. QPW_p - fixed ex-ante as option 1 is being used
4. $t_{p,y}$ - the project technology is not standardized flow rate technology and hence this parameter is not deemed applicable.
5. $P_{p,f,y}$ - fossil fuel consumption is not envisaged in the VPA.
6. LE_y - A % default leakage value has been considered.

SDG Indicator	SDG 13: Climate Change
Data / Parameter	$M_{q,y}$
Unit	fraction
Description	Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard of host country
Source of data	Field Test / lab Test results
Value(s) applied	0.90
Measurement methods and procedures	Water quality testing of the project devices on sampling basis. The samples of treated water collected from project devices, should be tested either using field testing kits or in labs.
Monitoring frequency	Annually or more frequently. In addition, the first round of testing will be conducted at least after six months from the start date of registration ²² .
QA/QC procedures	Following QA/QC procedures will be followed: 1. Laboratories used for water quality testing must be approved by local health authorities and/or have quality accreditation; and 2. The laboratory used must demonstrate that it has an adequate quality management plan in place which addresses both quality assurance and quality control test procedures.

²² The methodology mentions first test to be conducted within 6 months of project start date which is not possible for retroactive VPAs. Hence this has been altered to be conducted within 6 months of project registration.

	3. Field testing kits also are eligible, e.g. based on Colony Forming Unit method or Most Probable Number method. To use the field testing kits the project shall meet the following requirements: a. Testing kits must be approved by national agency or meet standards set by relevant international organization e.g. US-EPA, and b. Testing kits shall be tested for its accuracy and robustness prior to application for project level monitoring, whereby local or accredited laboratory shall conduct water quality tests using testing kits and a relevant ISO standard or an equivalent standard, in parallel with field testing kits. The sample size will be sufficient to meet 95/10 (or 90/10 as applicable) confidence/precision. The lower bound of the one-sided confidence interval may optionally be used as an alternative to the 95/10 (or 90/10 as applicable) rule.
Purpose of data	Determination of baseline emissions
Additional comment	If the proportion of samples not meeting Safe Drinking Water Quality Standards exceeds the following threshold, no emission reductions can be claimed for the corresponding monitoring period. Thresholds: - Project or VPA year 1: 20% - Project or VPA year 2: 15% - Project or VPA year 3 or above: 10% When the crediting period is renewed, the year number count continues, i.e. the second crediting period would encompass year 6, year 7, year 8, etc. Additionally, when the threshold is exceeded, the project shall provide an explanation for why this occurred and provide a remediation plan.

SDG Indicator	SDG 13: Climate Change
Data / Parameter	Water hygiene education campaigns
Unit	NA
Description	Hygiene campaigns carried out among project safe water end users
Source of data	Annual hygiene campaigns records
Value(s) applied	--

Measurement methods and procedures	Usage Survey (in person or telephone or messaging) A questionnaire-based survey will be used to assess hygienic handling of clean water as per CME knowledge and experience and WHO/UNICEF JMP core questions on drinking and hygiene. The PD shall report the activities conducted each year in a detailed "Report of annual hygiene campaigns results" and summarize the results in the project monitoring report Any significant change in the health status of the water users as a result of contaminated water (e.g. an outbreak of water related disease) must be reported.
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting.
Purpose of data	--
Additional comment	Sample size to be determined using 95/10 (or 90/10 as applicable) confidence precision requirements.

SDG Indicator	SDG 13: Climate Change
Data / Parameter	$X_{\text{cleanboil},y}$
Unit	Percentage
Description	Proportion of project end-users that boil safe (treated, or from safe supply) water after installation of project technology
Source of data	Project survey
Value(s) applied	0 %
Measurement methods and procedures	This survey may be performed in person, by telephone, by messaging (e.g. text, app), appropriate to the context.
Monitoring frequency	Annually
QA/QC procedures	Transparent data analysis and reporting is ensured through the use of personnel extensively trained in conducting Project Surveys. Results of the test will be reviewed by additional personnel trained in conducting the test. The remuneration of both the party undertaking the test and the party reviewing the result will not be linked to the performance of the VPA.
Purpose of data	Determination of baseline emissions
Additional comment	-

SDG Indicator	SDG 13: Climate Change
Data / Parameter	$HN_{p,y}$
Unit	Number

Description	Number of individuals per premises type p in the project boundary in year y
Source of data	Sales or distribution records
Value(s) applied	420
Measurement methods and procedures	The total number of individual for each project premise type in each specific VPA is tracked in the Sales or distribution records
Monitoring frequency	Continuous
QA/QC procedures	CME shall apply referenced cross-checks for Parameter $HN_{p,y}$. The value applied shall be cross-checked against at least one other sources like: - Project survey - Official government publications or statistics - Credible published literature for project region, or - Studies by academia, NGOs or multilateral institutions For cross-check purposes, sources applied may be up to 5 years old. Further, cross-check with older sources may be used provided they provide conservative results. Further, as per methodology clause 3.11.2, the source applied to cross-check the parameter may not be the most conservative, if it can be shown to be the most relevant. Refer Para 3.11.2 (a) and (b) of methodology for additional guidance.
Purpose of data	Determination of baseline emissions
Additional comment	-

SDG Indicator	SDG 13: Climate Change
Data / Parameter	$N_{p,y}$
Unit	Number
Description	Accumulated number of premises type p with at least one individual project technology in year y
Source of data	Sales or distribution records
Value(s) applied	650
Measurement methods and procedures	The total number of premises with at least one individual technology type is tracked in the Sales or distribution database.
Monitoring frequency	Annually
QA/QC procedures	Sales or distribution records to include: - Date of sale/distribution - Geographic area of sale - Model/type of project technology sold - Quantity of project technologies sold

	Name and telephone number, and address (if available) or other traceable indicator of premises identity and location for all end users
Purpose of data	Determination of baseline emissions
Additional comment	--

SDG Indicator	SDG 13: Climate Change
Data / Parameter	$U_{p,y}$
Unit	percentage
Description	Usage rate of the project technology by premises type p during year y
Source of data	Project Surveys
Value(s) applied	90%
Measurement methods and procedures	PP will use either In-person survey of project premises or by Survey performed by telephone or messaging or a mix of both options following Annex 1 of the applied GS methodology(non-binding).
Monitoring frequency	Annually
QA/QC procedures	--
Purpose of data	Determination of baseline emissions
Additional comment	The usage survey provides a single usage parameter that is representative for project technologies in the total sales record.

SDG Indicator	SDG 13: Climate Change
Data / Parameter	$DP_{p,y}$
Unit	Days
Description	Average days the project technology is present for end-users in the premises p in year y
Source of data	Sales and distribution records and School Academic Calendar
Value(s) applied	210
Measurement methods and procedures	Based on the sales or distribution records of "Date of sale/distribution" and ex-ante parameter "Expected technical life of project technology," determine for each project device how many days of the year it has been in the premises and within its technical life. Calculate the average for all the project technology by premises type p to obtain this parameter. For schools and other institutions, the days will be limited by the number of school days in the monitoring period, taking into account weekends and holidays, as applicable for boarding and non-boarding premises.

Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	Determination of baseline emissions
Additional comment	

SDG Indicator	SDG 13: Climate Change
Data / Parameter	$DN_{p,y}$
Unit	Number
Description	Average number of individual project technologies in each project premises type p in year y
Source of data	Sales or distribution records
Value(s) applied	Refer ex-ante ER sheet
Measurement methods and procedures	The total number of individual project units by each project premise type in each specific VPA is tracked in the Sales or distribution records
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	Determination of baseline emissions
Additional comment	-

B.7.2. Sampling plan

>>

Due to the large number of units projected to be distributed in the VPA, coupled with the difficulties with monitoring in some regions, it is not economically feasible to monitor each unit distributed in the VPA. Therefore, the project will employ representative sampling in line with the requirements of the applied methodology.

Stratified random sampling shall be applied for determining the usage rate of units distributed. The units in the population shall be categorized (based on technology deemed appropriate) and sample size shall be determined based on expected usage rate in each category using stratified random sampling approach as per the Guideline: Sampling and surveys for CDM project activities and programmes of activities and Standard "Sampling and surveys for CDM project activities and programmes of activities". The sample size determined shall be distributed within each category based on percentage of units in corresponding category. A deviation (from monitoring methodology) on the sampling approach has been approved by GS4GG for the first crediting period. Please refer Appendix 5.

Baseline / Project surveys, if any, should be carried out as per the sample sizing mentioned in para 4.2.3, page 41 of the applied methodology however, the sampling exemptions stated in Appendix 5 below shall prevail.

Sampling approaches / plans may be updated during the crediting period in case they are deemed better/convenient than stratified random sampling.

In some situations, the project proponent reserves the right to conduct monitoring remotely (like interviews by phone, message or remotely through its staff / trained local personal to assess parameters that will not need to be physically measured). Alternatively, the project proponent will establish conservative values based on data from the past verifications and/or from local knowledge from the in-country team in case monitoring of some parameters is not possible either via physical or remote monitoring.

Conservative bound value of 95/10 or 90/10 as applicable shall be used in case the desired precision is not achieved instead of repeating the monitoring.

B.7.3. Other elements of monitoring plan

>>

Implementation Plan:

Impact Carbon is responsible for the production of monitoring reports, following the criteria outlined in section B.7.1 and B.7.2. Sampling will be carried out following the Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities in accordance with GS4GG requirements. A precision/confidence of 95/10 will be applied for cross-VPA sampling, in accordance with the requirements of the methodologies above.

Sampling efforts will be conducted by qualified personnel or those who have prior experience of sampling. The surveyors will have native language(s) competencies, allowing for full comprehension of any responses given by sampled users, and any questions therein.

Data collection and archiving:

The monitored data will be entered into the CME database. Original copies of monitoring documents will be kept and archived. The name of the surveyor will be specified on monitoring survey forms, therefore allowing for the follow-up of incomplete/unclear data, in case any. Hard copies of all documents will be kept at the office of Impact Carbon, or an alternative address nominated by them, and monitoring data will be entered into a central record keeping database. The record keeping database will be used to record the results of all monitoring/total sales.

The CME shall maintain the following information:

- 1) **Total sales record:** The total sales record contains the information listed below for the technologies implemented under the project. This record will be kept electronically and/or in paper records (wherever applicable and possible) and provided to DOE at verification. Records will include the following information:
 - a. Type of system (UltraFLO / UltraTAB)
 - b. Unique serial number of the units installed / distributed
 - c. Date of installation / distribution
 - d. Address and details of school and contact detail (if available) of representative
 - e. Type of School (Boarding / Non-boarding)
 - f. Number of project technology installed/distributed in an institution
 - g. School population count (number of students / staff in boarding / non-boarding category)

- 2) The project surveys will collect information as per the survey questionnaire designed for the purpose.

Responsibilities and institutional arrangements:

The responsibility for monitoring and reporting lies with the CME. Trained staff will be assigned to carry out the monitoring process including data recording, reporting, archiving and management.

Entities responsible for conducting monitoring

Role	Responsibility
Project Development Director	Receive escalated monitoring issues and questions from Water Programme Manager, clarify uncertainties in methodology, provide additional support from Project Development team if needed. Advises on monitoring issues.
Programme Manager -	- Enlist and manage work of Third-party monitoring - supply protocols, ensure timeframe is met, address issues, compile final reports, etc - Manages the Project Database, in which the results of monitoring shall be summarized - Oversees and drafts the production of annual monitoring reports - Coordinates communication with the verifier and Gold Standard.
VPA Implementer	- Collecting data to be monitored accurately, or training Field Measurement Personnel to do so. - Sharing monitoring data with the Water Programme Manager (CME). - Maintains proper and continuous records of project activities and disseminated WPS, including technology ident
Programme Associate	Assist with the completion of monitoring reports with input from the Water Programme Manager
Field measurement personnel	Conduct on the ground monitoring of sampled users
External QAQC	- Verify the monitoring work done to ensure accuracy before submission. - review protocols, interview enumerators, spot check data

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

>>

12/01/2018

C.1.2. Expected operational lifetime of project

>>

21 Years and 0 Months

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>

The start date of the First crediting period is 01/01/2021

C.2.2. Total length of crediting period

>>

Total length of the crediting period is 15 years (5 years renewable twice). The length of the first crediting period is 5 years which is from 01/01/2021 to 31/12/2025.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 1. Human Rights	not required
Principle 2. Gender Equality	not required
Principle 3. Community Health, Safety and Working Conditions	not required
Principle 4.1 Sites of Cultural and Historical Heritage	not required
Principle 4.2 Forced Eviction and Displacement	not required
Principle 4.3 Land Tenure and Other Rights	not required
Principle 4.4 Indigenous people	not required
Principle 5. Corruption	not required
Principle 6.1 Labour Rights	not required
Principle 6.2 Negative Economic Consequences	not required
Principle 7.1 Emissions	not required
Principle 7.2 Energy Supply	not required
Principle 8.1 Impact on Natural Water Patterns/Flows	not required
Principle 8.2 Erosion and/or Water Body Instability	not required

Principle 9.1 Landscape Modification and Soil	not required
Principle 9.2 Vulnerability to Natural Disaster	not required
Principle 9.3 Genetic Resources	not required
Principle 9.4 Release of pollutants	not required
Principle 9.5 Hazardous and Non-hazardous Waste	not required
Principle 9.6 Pesticides & Fertilisers	not required
Principle 9.7 Harvesting of Forests	not required
Principle 9.8 Food	not required
Principle 9.9 Animal husbandry	not required
Principle 9.10 High Conservation Value Areas and Critical Habitats	not required
Principle 9.11 Endangered Species	not required

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Requirement	Relevance	Description	Mitigation Action Required
a. Is there a possibility that the VPA might reduce or put at risk women's access to or control of resources, entitlements and benefits?	Yes	The Project does not limit women access to or control of resources, entitlements and benefits. On the contrary, the Project ameliorates the living condition of women and children by freeing them from the burden of biomass fuel collection, decreasing smoke levels and incidence of respiratory illness relative to use of traditional biomass stoves for water boiling	Not required
b. Is there a possibility that the VPA can adversely affect men and women in marginalised or vulnerable communities (e.g., potential increased burden on women or social isolation of men)?	Yes	The Project does not result in any increased burden on women or social isolation of men. The Project ameliorates the living condition of women by freeing them from the burden of biomass fuel collection as well as saving water boiling time.	Not required

c. Is there a possibility that the VPA might not take into account gender roles and the abilities of women or men to participate in the decisions / designs of the project's activities (such as lack of time, child-care duties, low literacy or educational levels, or societal discrimination)?	Yes	The Project ameliorates the living condition of women by freeing them from the burden of fuelwood collection and freeing them from water boiling time. Any time savings will be used by the women for more productive work or child-care duties.	Not required
d. Does the PoA take into account gender roles and the abilities of women or men to benefit from the Project's activities (e.g., Does the PoA / VPA criteria ensure that it includes minority groups or landless peoples)?	No	The Project does not de-limits its access to minority groups or landless people	Not required
e. Does the VPA design contribute to an increase in women's workload that adds to their care responsibilities or that prevents them from engaging in other activities	Yes	The Project reduces women's workload by freeing them from the burden of fuelwood collection and reduce water boiling time, allowing more time for alternative activities	Not required
f. Would the VPA potentially reproduce or further deepen discrimination against women based on gender, for instance, regarding their full participation in design and implementation or access to opportunities and benefits?	Yes	The Project is not expected to reproduce / deepen discrimination against women. In fact as listed above, the Project offers multiple benefits and opportunities to women	Not required
g. Would the VPA potentially limit women's ability to use, develop and protect	Yes	The Project is not expected to limit women' ability to use, develop or protect natural resources. In fact as	Not required

natural resources, taking into account different roles and priorities of women and men in accessing and managing environmental goods and services?		listed above, the Project offers multiple benefits and opportunities to women to avoid deforestation resulting in protection of forests	
h. Is there a likelihood that the proposed VPA would expose women and girls to further risks or hazards?	Yes	The Project does not expose women or girls to further risks or hazards. By providing access to clean and safe cooking technology, the Project reduces risk to health due to IAP.	Not required

The VPA seeks gender certification at first level only (Mandatory Gender-Sensitive Requirements). Thus, the VPA needs to demonstrate compliance only with step 1-3 of the GS4GG Gender Equality Requirements & Guidelines. Please note the following in this regard:

- The project reflects the key issues and requirements of Gender Sensitive design and implementation as per GS Gender Policy. The CME has put in place internal policies that are gender sensitive and ensure that equal opportunities / access is available to women and men alike on various issues. The project is non-discriminating on the basis of gender in addition to other non-discriminating parameters. The project provides equal opportunity to men and women with respect to access to its technology / measures and use of resource, two-way stakeholder communication, voicing grievances, employment etc.
- The following table details the project's alignment with nation policies, strategies and best practices:

National Policy	Project alignment
In Nigeria "committee on the Elimination of Discrimination against Women Thirtieth session 12-30 January 2004" raised its concerned about the continuing prevalence of violence against women and girls, including domestic violence and sexual harassment in the workplace	Impact Carbon, as part of HR policy has zero tolerance on issues around sexual harassment and misconduct. The Policy defines the scope, coverage, procedure for complaints and punishment in such cases. The contact details of Internal Committee set up to review such concerns, are listed on the notice boards and are equally accessible to all genders.
In the year 2000, Nigeria adopted and passed into law the National Policy on Women. The need for the policy was "to fulfill the yearnings as well as (promote) the efforts of Federal, State and Local Governments, Non-	The project provides equal opportunities to women and men alike and does not result in any increased burden on women or social isolation of women.

governmental Organizations, International Development Partners, the private sector, concerned corporate bodies and individuals to integrate women fully into national development, in order to remove those gender inequalities that have evolved through structures and processes created by patriarchy, colonialism and capitalism”.	
President of Nigeria declared a State of Emergency in 2018 and launched the National Action Plan (NAP), a 13-year strategy for the Revitalization of Nigeria’s Water Supply, Sanitation, and Hygiene (WASH) Sector aimed at ensuring universal access to sustainable and safely managed WASH services by 2030, commensurate with the Sustainable Development Goals. Women and girls suffer disproportionately from the lack of adequate WASH services. They bear the burden of water collection over long distances, which has been associated with negative effects on well-being, school attendance, and a higher risk of gender-based violence (GBV). Access to WASH can impact years of schooling by freeing up time that children spend collecting water to attend school, reducing the prevalence of disease that can keep them out of school, and contributing to a safe and healthy learning environment while at school.	The project increases women access to or control of resources, entitlements and benefits. The project by installation/distribution of WPS units upgrades the living condition of women by freeing them from the drudgery of water collection over long distance and fuel collection for water boiling on inefficient stoves and reduces the impact of IAP and water borne diseases on health, this results in more number of school days for students. Schools/institutions, where the WPS units are installed, does not discriminate on basis of gender with respect to access to the safe drinking water.

- The information sharing under the project is equitable with women and men stakeholders as a minimum standard. The information related to the project available and presented in an accessible format across all stakeholder groups. This was ensured via sharing non-technical summary via email to stakeholders without discriminating on the basis of gender. The project provides equal opportunities to all stakeholders to voice their comments /concerns. There are opportunities for stakeholders to share information in a two -way exchange, give regular feedback during implementation and ensure their views and priorities are incorporated in design and practice. The project has set up a comprehensive grievance mechanism system which is accessible to women and men alike.

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	Yes The project is aligned with existing country policies, strategies and best practices The project addresses the Gender sensitive requirements as mentioned in table above
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The project complies with the National Gender Policies as explained above
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No, as explained in tables above, the project has significant positive contribution towards gender equality. Further the consultation conducted for the VPAs are in line with Gold Standard Stakeholder Consultation & Engagement Procedure, Requirements & Guidelines. The invitations extended for the consultation conducted already included Ministry of Education, Ministry of Health, Ministry of Environment, National Environment Management Agency, UNWOMEN to provide their comments/ feedback. Refer GS4GG SCR wherein the stakeholders' have provided positive rating to the project towards SDG5. Based on above, an expert's assistance on Gender/Environment issues is not deemed required
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No, refer above

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

For VPA level consultation, clause 6.1.4 of the GS4GG Programme of Activity Requirements" V1.2, requires minimum two rounds of consultation including one mandatory physical meeting and one stakeholder feedback round. Further as per clause 6.1.7, for retroactive VPA/CPA, a physical meeting shall be held at the stakeholder feedback round if no physical meeting has taken place earlier.

The systems under the concerned VPA have been transferred from the CPAs under CDM PoA 9948. As part of CDM PoA 9948, stakeholder consultation meetings were undertaken at CPA level (covering a group of CPAs). The physical Local Stakeholder

Consultation meetings were held on 14th March 2019 at 09:00 A.M. at the Best Western Hotel in Lagos, Nigeria²³ for WPS systems installed in Nigeria. This is verifiable from the registered CPA-DDs under CDM PoA 9948. Thus, given a physical meeting has already been conducted for these CPAs under CDM PoA 9948, hence as per clause 6.1.7, physical meeting again for these CPAs is not deemed required.

Therefore, this VPA created via transitioning systems from CDM PoA 9948 is compliant with the requirements related to stakeholder consultation and do not require any additional physical stakeholder meeting.

Considering aforesaid, the CME conducted VPA level stakeholder consultation (SFR), covering the identified gaps between CDM and GS4GG requirements for stakeholder consultation.

The SFR was initiated on May 14, 2021, it was due for closure on July 13, 2021. However, due to COVID, the CME extended the consultation time to provide equal opportunity to various stakeholders to revert. Hence, instead of the required 60 days period, the SFR feedback window had been kept open for more than 100 days and was concluded on Tuesday, August 31, 2021.

Lastly, CME conducted a virtual stakeholder consultation meeting also on Tuesday, August 31, 2021, Via Google Meet. The entire meeting session was recorded online. For more detail, refer to the GS4GG Stakeholder Consultation Report.

E.1. Summary of stakeholder mitigation measures

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During the SFR conducted, in total 33 stakeholders responded to the feedback request (14 for Nigeria and 19 for Kenya) attributed to the comprehensive follow up efforts, reminders raised by CME and extended duration of SFR. In general, the stakeholders were appreciative of the project and its features and acknowledged the reduction in water borne diseases, Indoor air pollution and GHG emissions, employment generation and other SDG impacts of the project. The stakeholders liked the fact that the VPA sought detailed and comprehensive feedback on various features of the VPAs.

For details refer SFR feedback forms provided under Annex 1 of GS4GG Stakeholder Consultation report for details.

No mitigation action required.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.	Justification
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²³ https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/O4BIKAVRLH7Y2QE61FPUJWNTX8SG5Z/view

Continuous Input / Grievance Expression Process Book (mandatory)	Continuous input / Grievance Expression process book is available at the office at the following address: Impact Water Global Impact H2O Innovations, Ltd. Plot 61, Adekunle Fajuyi Street, G.R.A. Ikeja, Lagos Nigeria.	In line with section 2.1 of the Annex W Expression book has been placed at office of Impact Water in Nigeria. Stakeholders are free to voice their concerns via the Grievance Expression Book. By maintaining feedback book at the local office, it is ensured that stakeholders that don't have access to electronic media for expressing concerns / grievances are also able to share their concerns / feedback. Additionally, the end users always have an option to revert to the salesperson (representative of distribution/retail partners etc.) in case of any feedback / complaints with the product post distribution.
GS Contact (mandatory)	help@goldstandard.org	--
Other	Contact number: +256 790 911 934	As the project is spread across a huge area hence telephone access has also been provisioned for, in line with Annex W, section 2.3 of Gold Standard.
	Email: info@impactcarbon.org	As per para 2.4 of Annex W of GS, the stakeholders with internet access have an option of contacting Impact Carbon through the email id provided.

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

The assessment of Safeguarding Principles is carried out in line with the "Gold Standard for the Global Goals Safeguarding Principles and Requirements" version 1.2, published in October 2019 available at the following link:

<https://globalgoals.goldstandard.org/103-par-safeguarding-principles-requirements/>

The assessment is as follows:

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/ no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measure s added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	Yes	The VPA and CME both respect human rights and are not complicit in violence or human rights abuses.	Not required
2. The Project shall not discriminate with regards to participation and inclusion	Yes	The VPA does not discriminate with regards to participation and inclusion	Not required
Principle 2. Gender Equality and Women's Rights			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women a. Sexual harassment and/or any forms of violence against women – address the multiple risks of gender-based violence, including sexual exploitation or human trafficking.	No	Not relevant	Not required

b. Slavery, imprisonment, physical and mental drudgery, punishment or coercion of women and girls.	No	Not relevant	Not required
c. Restriction of women's rights or access to resources (natural or economic).	No	Not relevant	Not required
d. Recognise women's ownership rights regardless of marital status – adopt project measures where possible to support to women's access to inherit and own land, homes, and other assets or natural resources.	No	Not relevant	Not required
2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work a. Where appropriate for the implementation of a PoA/VPA, paid, volunteer work or community contributions will be organised to provide the conditions for equitable participation of men and women in the identified tasks/activities.	No	Not relevant	Not required
b. Introduce conditions that ensure the participation of women or men in Project activities and benefits based on pregnancy, maternity/paternity leave, or marital status.	No	Not relevant	Not required
c. Ensure that these conditions do not limit the access of women or men, as the case may be, to PoA/VPA participation and benefits.	No	Not relevant	Not required
3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks	Yes	No gender risks are envisaged in the VPA	Not required
4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	Not relevant	Not required
Principle 3. Community Health, Safety and Working Conditions			

1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	Yes	The VPA reduces exposure to indoor air pollutants and smoke levels, further reducing incidence of respiratory illness compared to cooking/water boiling on traditional biomass stoves using solid biomass fuel/fossil fuels. The VPA also reduce incidence of diarrhoea and water borne diseases etc. due to drinking of unsafe water.	Not required
Principle 4 Cultural Heritage, Indigenous Peoples, Displacement and Resettlement			
Principle 4.1 Sites of Cultural and Historical Heritage			
1. Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	Not relevant	Not required
Principle 4.2 Forced Eviction and Displacement			
1. Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	Not relevant	Not required
Principle 4.3 Land Tenure and Other Rights			
1. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	Not relevant	Not required
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?"	No	Not relevant	Not required
Principle 5. Corruption			

1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	Yes	The CME does not promote / or is complicit in direct or indirect corruption.	Not required
Principle 6 Economic Impacts			
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions	Yes	The VPA does not involve any forced labour and the PP ensures that all employment is in compliance with local labour regulations and laws.	Not required
2. Workers shall be able to establish and join labour organisations	Yes	The CME puts no constraints / limitation on employees to form a union.	Not required
3. Working agreements with all individual workers shall be documented and implemented and include: a. Working hours (must not exceed 48 hours per week on a regular basis), AND b. Duties and tasks, AND c. Remuneration (must include provision for payment of overtime), AND d. Modalities on health insurance, AND e. Modalities on termination of the contract with provision for voluntary resignation by employee, AND f. Provision for annual leave of not less than 10 days per year, not including sick and casual leave.	Yes	The CME's policies and employment contracts are compliant with the requirement	Not required
4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion)	No	The CME does not promote / or is complicit in child labour	Not required
5. The Project Developer shall ensure the use of appropriate equipment, training of workers,	No	Not relevant	Not required

documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	Yes	No negative economic consequences are deemed applicable	Not required
Principle 7 Climate and Energy			
Principle 7.1 Emissions			
1. Will the Project increase greenhouse gas emissions over the Baseline Scenario?	Yes	The VPA reduces GHG emissions relative to baseline scenario	Not required
Principle 7.2 Energy Supply			
1. Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	Yes	The project will reduce fuel resource consumption	Not required
Principle 8 Water			
Principle 8.1 Impact on Natural Water Patterns/Flows			
1. Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	Not applicable	Not required
Principle 8.2 Erosion and/or Water Body Instability			
1. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	Yes	The VPA shall result in reduction in demand of biomass fuel / fossil fuel in the region putting less pressure of forests for deforestation and will hence indirectly avoid erosion associated with tree cutting/ felling.	Not required
Principle 9 Environment, Ecology and Land Use			

Principle 9.1 Landscape Modification and Soil			
1. Does the Project involve the use of land and soil for production of crops or other products?	No	Not applicable	Not required
Principle 9.2 Vulnerability to Natural Disaster			
1. Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	Not applicable	Not required
Principle 9.3 Genetic Resources			
1. Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	Not applicable	Not required
Principle 9.4 Release of pollutants			
1. Could the Project potentially result in the release of pollutants to the environment?	No	Not applicable	Not required
Principle 9.5 Hazardous and Non-hazardous Waste			
1. Will the Project involve the manufacture, trade, release, and/or use of hazardous and non-hazardous chemicals and/or materials?	No	Not applicable	Not required
Principle 9.6 Pesticides & Fertilisers			
1. Will the Project involve the application of pesticides and/or fertilisers?	No	Not applicable	Not required
Principle 9.7 Harvesting of Forests			
1. Will the Project involve the harvesting of forests?	Yes	The VPA does not involve harvesting of forests. The PoA shall result in reduction in demand of biomass fuel / fossil fuel in the region putting less pressure of forests	Not required

		for deforestation and will hence indirectly avoid erosion associated with tree cutting/ felling.	
Principle 9.8 Food			
1. Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	Not applicable	Not required
Principle 9.9 Animal husbandry			
1. Will the Project involve animal husbandry?	No	Not applicable	Not required
Principle 9.10 High Conservation Value Areas and Critical Habitats			
1. Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	Not applicable	Not required
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	Not applicable	Not required
AND/OR			
Does the Project potentially impact other areas where endangered species may be present through transboundary affects?			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	Impact Carbon
Registration number with relevant authority	-
Street/P.O. Box	582 Market St, Suite 1205
Building	
City	San Francisco
State/Region	CA
Postcode	94104
Country	USA
Telephone	+1 (415) 968-9087
E-mail	ehaigler@impactcarbon.org
Website	www.impactcarbon.org
Contact person	Evan Haigler
Title	Director
Salutation	Mr.
Last name	Haigler
Middle name	
First name	Evan
Department	Executive
Mobile	
Direct tel.	
Personal e-mail	

APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

N/A

APPENDIX 4-ENTITY RESPONSIBLE FOR COMPLETING VPA DESIGN DOCUMENT

Project participant and/or responsible person/ entity	☐ Project participant ☒ Person/entity responsible for completing the VPA-DD
Organization name	Climate Secure India Private Limited
Street/P.O. Box	Club Road
Building	Pragati Apartments

City	West Delhi
State/Region	Delhi
Postcode	110063
Country	India
Telephone	+91 11 2521 3080
Fax	--
E-mail	info@climate-secure.com
Website	www.climate-secure.com
Contact person	Rohit Lohia
Title	Director
Salutation	Mr.
Last name	Lohia
Middle name	--
First name	Rohit
Department	--

APPENDIX 5- DEVIATION FROM THE MONITORING METHODOLOGY

Reference: Dev_184

Date of approval: 07/12/2021

Validity: first crediting period of the VPA

Description:

Methodological requirement:

- 1. Parameter table SDWS 29 - The minimum sample size for IWT – for individual technology age group shall be determined considering the project technology type and in line with the sampling approach applied*
- 2. Also, page 51 of the applied methodology, for parameter SDWS29, refers a minimum sample size of 30 per technology age.*
- 3. Para 4.2.2. of the applied methodology prohibits grouping more than 10 VPAs together.*

Deviation proposed:

Conduct the project surveys for WPS stratifying each technology irrespective of their age, and by grouping more than 10 small scale VPAs together.

Justification:

WPS technologies are not affected by ageing, and are resupplied, maintained, and/or replaced on an ongoing basis. The water purification technologies operate

on consumable modules basis i.e., once their treatment capacity (cartridge / tablets / filters) is fully consumed, their consumables (cartridges / tablets / filters) are replaced making them revive their useful lifetime (age) again. Also, WPS operate on binary performance rather than reducing performance i.e., a WPS irrespective of its age will either provide safe water or unsafe water. Thus, requirement to monitor the systems per technology age is deemed superfluous and only the monitoring of each WPS technology, irrespective of age shall be applicable for WPS devices.

The VPAs in each country are identical as they follow same management plan, operational plan and technologies. The limit on grouping by number of VPAs (maximum 10) is not a correct parameter to determine the limits for sampling.

GS Decision:

The deviation request is approved for the first crediting period of the VPAs GS11259 to GS11305. However, the PD must ensure that:

1. The stratified random sampling approach applied for sample size calculation for the monitoring survey must meet all the requirements under Appendix 3 of Guideline: Sampling and surveys for CDM project activities and programmes of activities.
2. Continuity in the project's monitoring activities is maintained, and PD can justify that no monitoring gaps exist (especially for SDG parameters) within the Monitoring Period(s). However, if gap(s) exist, the project shall justify that conservative approach(es) have been applied in line with section 3 of the Deviation Approval Requirements and Procedures (version 1.1) and overarching GS principles (as applicable).

Revision History

Version	Date	Remarks
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption